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POPULATION DYNAMICS AND LIFE HISTORIES OF FOLIACEOUS CORALS¹

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Abstract. The population dynamics of five species of foliaceous corals (*Agaricia agaricites* forma *purpurea*, *A. lamarcki*, *Leptoseris cucullata*, *Montastrea annularis*, and *Porites astreoides*) was followed on Jamaican reefs using annual photographic censuses. Overall, population cover, size frequencies, and number of colonies were stable over the monitored period from 1977 to 1980. However, individual colonies were in turmoil: of the original 883 colonies, 315 were killed outright and 499 suffered partial colony mortality (injury) at least once during the 3 yr. Partial mortality generated an additional 189 colonies by fission, while larval recruitment added another 201, and fusion subtracted 40 colonies. The net result was a decrease of <10% in number of colonies. There was considerable variation among years and sites in measured life history parameters, as well as striking differences between species. The most stable populations were *M. annularis* and *A. lamarcki*, followed by *P. astreoides*, *A. agaricites*, and *L. cucullata*.

Rates of partial- and whole-colony mortality were strongly dependent on colony size for all species. Typically, small colonies either were unharmed, or were killed outright, while most large colonies survived but were injured each year, often by extensive amounts. The amount of tissue lost from a population through injuries was usually much greater than through the death of whole colonies, even in a year which included a major winter storm. Frequently, large corals were split asunder by partial mortality to produce several daughter colonies, which presumably were of identical genotype. Therefore counts of physically separate colonies exceeded the number of genetically distinct individuals (genets), by at least 20%. Individual genets, measured as the lateral extent of known daughter colonies, were frequently up to 5 m across, and for *M. annularis* and *A. lamarcki* were certainly several centuries old.

Colony extension rates measured in situ were very weakly dependent on depth from –10 to –55 m, and were independent of colony size. Small colonies showed much faster relative changes in area, although even the largest corals continued to grow if they avoided major injuries. Within a size-class, the fates of colonies were diverse because of differential rates of growth and shrinkage, so that size was a very poor indicator of age.

Differences in the life history and “mobility” between species are reflected in the taxonomic and morphological composition of coral communities over the reef. Shallow-water assemblages of foliaceous corals are composed of more dynamic, delicately built species, while many deeper water communities are dominated by slower growing, robust species. Ironically, disturbance on coral reefs often seems to favor the organisms most vulnerable to damage.

Key words: clones; coral reefs; demography; disturbance; fission; fusion; genet; historical effects; life histories; module; partial mortality; population dynamics.

INTRODUCTION

The demographic traits of organisms play a central role in their ecology and evolution (Stearns 1976, 1977). Despite their importance, quantitative estimates of life history parameters are comparatively rare and highly biased towards organisms that are solitary, mobile, short-lived, or of commercial importance (e.g., *Homo*, *Drosophila*, *Tribolium*). In particular, the population biology of clonal animals and plants has been relatively neglected in spite of their obvious ecological impor-

tance. (For a summary of recent work see Jackson et al. 1985.) For example, Connell (1973:221) noted in a comprehensive review that there had not been a single population study of a scleractinian coral that incorporated a wide range of colony sizes and a large number of replicates. Even with the enormous interest of coral reefs to theoreticians and community ecologists over the last century (e.g., Darwin 1842, Slobodkin and Sanders 1969, Connell and Slatyer 1977), detailed, species-level demographic studies of subtidal coral populations are presently limited to a single weedy species (Loya 1976a, b, Rinkevich and Loya 1979a, b).

Despite our ignorance of coral demography, evidence is mounting that coral reef communities are extremely dynamic (e.g., Connell 1973, 1976, Lewis 1974, Loya 1976a, b, Birkeland 1977, Dustan 1977, Bak and

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Luckhurst 1980, Karlson 1980, Porter et al. 1981), especially during recovery from storms or other disasters (Glynn et al. 1964, Shinn 1972, Loya 1976c, Bak and Crieis 1981, Colgan 1981, Knowlton et al. 1981, Pearson 1981, Woodley et al. 1981, and many more). Earlier we have shown that even though rates of recruitment, growth, and mortality of a common Caribbean coral, *Agaricia agaricites* (Linnaeus) were high, the population structure remained very stable over the course of a year (Hughes and Jackson 1980). In the present paper we test the generality of those preliminary observations for a range of species, between sites and over several years. This allows us for the first time to compare in detail the population dynamics and life histories of shallow- and deep-water coral species. These measurements of spatial and temporal variability in demographic parameters are an essential first step towards an understanding of how coral communities are organized (cf. Connell and Slatyer 1977).

Colonial animals such as most corals, bryozoans, and ascidians grow by clonal iteration (budding) of repeated building blocks analogous to individual solitary organisms (Beklemishev 1969, Boardman et al. 1973, Jackson 1977). These modules (e.g., coral polyps, bryozoan zooids) can usually function and survive alone or in small groups when isolated from the rest of the colony, and are thus equivalent to ramets of plant ecologists (Harper 1977, Cook 1985). The sum of all ramets derived from the same zygote, whether situated in one or several separate colonies, makes up the genet. This mode of growth of clonal organisms imposes on them a suite of demographic characteristics not usually found among other acclonal organisms. After budding, modules grow until they reach a maximum size characteristic of the species. The colony may continue to grow, however, by further budding of new units. Individual modules may die while others remain alive. Therefore, the balance between rates of module budding and death defines net colony growth, much like the role of births and deaths in the dynamics of populations of acclonal individuals (see review in Harper 1977).

In this paper we describe in detail the population dynamics of five species of colonial corals. We begin by documenting the numbers and sizes of colonies through time. Second, we examine processes which interact to determine population size (polyp and colony mortality, colony fission and fusion, and sexual recruitment). Third, we follow the growth of individual colonies to discover (1) how population and community structure are maintained, and (2) if demographic differences arise between corals that are originally the same size. Throughout the paper, we contrast the fates of polyps and colonies through time, and then relate these fates to the more general problem of characterizing the population dynamics of clonal organisms. We conclude by comparing the five coral species, and by relating their contrasting life histories to differences in distribution on Caribbean reefs.

STUDY ANIMALS, SITES, AND METHODS

Animals

Corals grow in a variety of forms, but we have limited our study to foliaceous (plating) species, which comprise the predominant morphology at most of our sites. We chose five species for study: *Agaricia agaricites* forma *purpurea*, *Agaricia lamarcki*, *Leptoseris* (formerly *Helioseris*) *cucullata*, *Montastrea annularis*, and *Porites astreoides*. All of these species are very common or abundant on the Jamaican reefs where this study was done, with extensive depth ranges from <3 m to >70 m (Goreau and Wells 1967). *Leptoseris cucullata* and *A. lamarcki* are foliaceous over their entire depth range, but *M. annularis*, *A. agaricites*, and *P. astreoides* are polymorphic, typically changing shape from mound-like to foliaceous with increasing depth. On most reefs in Jamaica, plating corals are common below ≈ 20 m, becoming the most abundant growth form from ≈ 30 to 70 m (Goreau and Wells 1967, Kinzie 1973). We stress that only foliaceous forms of morphologically variable species are present at our study sites, and we would expect other growth forms to have very different demographic properties from those described here (see Connell 1973, Highsmith 1982).

Sites

Pinnacle 1.—The first study area, known as Pinnacle 1, lies just offshore from Discovery Bay at a depth of 35 m. As the name suggests, the Pinnacle is a small (50×20 m) steep-sided reef, ranging in depth from -26 to -38 m. Almost all the corals present are foliaceous (Table 1, 35 m). The most conspicuous of these are *M. annularis*, *A. lamarcki*, *A. agaricites*, and *L. cucullata*. In 1977, 12 1-m² quadrats made of polyvinyl chloride waterpipe were tied with nylon rope onto the Pinnacle to mark permanently the position of large numbers of these four species. Sites of the quadrats were first located haphazardly, and then the quadrats were carefully positioned to avoid contact with the corals before tying down. The coral populations thus enclosed (386 colonies in July 1977) were monitored photographically each summer until 1980. Colonies with >50% of their surface area lying outside the quadrats in 1977 were not considered.

Deep Mounds.—Since we expected demographic patterns to vary as a function of depth, we tried to maximize the depth range of our investigations. Our second study area was situated directly seaward of Pinnacle 1 at a depth of 55 m. Corals at this site are confined to small rocky mounds, ranging in size from ≈ 1 –3 m³ (Goreau and Land 1974:Fig. 2). Because of the depth and topography, we limited our investigations to a study of colony growth rates using in situ staining (see below). The Deep Mounds are densely covered with foliaceous colonies of *M. annularis*, *M. cavernosa*, *A. lamarcki*, *A. grahamae*, *A. fragilis*, and *A. undata*. Ten of these small mounds were chosen

TABLE 1. Percentage cover (mean $\pm$ standard deviation) of major taxa at Rio Bueno (–10, –15, –20 m) and at Pinnacle 1 (–35 m). Data are from 10 10-m-long chain transects, run haphazardly at each depth in July 1977.

Taxa	Depth (m)			
	10	15	20	35
Fleshy algae	0	0.72 $\pm$ 0.78	0.06 $\pm$ 0.22	4.16 $\pm$ 6.24
Sponges	0.27 $\pm$ 0.28	5.52 $\pm$ 4.59	6.88 $\pm$ 7.11	8.19 $\pm$ 10.12
Branching corals				
<i>Madracis mirabilis</i> (Duchassaing & Michelotti)	16.93 $\pm$ 11.75	17.11 $\pm$ 10.93	4.27 $\pm$ 7.28	0
<i>Acropora cervicornis</i> (Lamarck)	6.00 $\pm$ 3.40	0.45 $\pm$ 0.80	0	0
<i>Porites furcata</i> Lamarck	10.12 $\pm$ 3.95	1.14 $\pm$ 1.38	0.25 $\pm$ 0.79	0
Massive corals				
<i>Colpophyllia natans</i> (Muller)	3.29 $\pm$ 5.32	1.57 $\pm$ 1.92	2.69 $\pm$ 7.42	0.50 $\pm$ 1.34
<i>Mycetophyllia</i> spp.	0.16 $\pm$ 0.42	0.29 $\pm$ 0.54	0.18 $\pm$ 0.39	0.89 $\pm$ 1.06
<i>Montastrea cavernosa</i> (Linnaeus)	0.12 $\pm$ 0.36	4.55 $\pm$ 8.44	2.33 $\pm$ 4.92	3.11 $\pm$ 5.64
<i>Siderastrea siderea</i> (Ellis & Solander)	0	4.28 $\pm$ 5.22	3.05 $\pm$ 3.19	0.29 $\pm$ 0.55
Plating corals				
<i>Agaricia agaricites</i> (Linnaeus)	16.28 $\pm$ 10.96	10.64 $\pm$ 5.73	8.26 $\pm$ 5.33	4.14 $\pm$ 3.92
<i>Agaricia lamarcki</i> Milne Edwards & Haime	0	2.17 $\pm$ 5.21	12.84 $\pm$ 15.86	12.08 $\pm$ 14.53
<i>Leptoseris cucullata</i> (Ellis & Solander)	1.53 $\pm$ 1.87	1.47 $\pm$ 1.80	2.05 $\pm$ 1.90	6.03 $\pm$ 5.80
<i>Porites astreoides</i> Lesueur	10.82 $\pm$ 8.70	5.87 $\pm$ 3.48	2.39 $\pm$ 2.92	0.11 $\pm$ 0.45
<i>Montastrea annularis</i> (Ellis & Solander)	2.36 $\pm$ 2.81*	1.28 $\pm$ 2.93*	1.05 $\pm$ 1.70	14.06 $\pm$ 12.18
Other coelenterates	2.14 $\pm$ 3.98	3.78 $\pm$ 5.65	1.11 $\pm$ 3.42	2.13 $\pm$ 3.46
Bare space/crustose algae/sediment	29.75 $\pm$ 7.83	40.07 $\pm$ 7.20	52.06 $\pm$ 16.55	43.77 $\pm$ 8.27
Total	99.77	99.91	99.47	99.46

* *M. annularis* colonies at –10 and –15 m were a mixture of plate- and mound-like morphologies.

haphazardly and marked with submerged buoys so they could be relocated each year.

Rio Bueno.—The third study site is 4 km west of Pinnacle 1, near the top of a vertical cliff on the west side of Rio Bueno Harbour. Here foliaceous corals are common as shallow as –8 m (Woodley and Robinson 1977, Jackson and Winston 1982). Twelve 1-m² quadrats identical to those at Pinnacle 1 were tied onto the reef wall at –10 to –20 m. The specific sites were chosen to contain large numbers of platey colonies of *A. agaricites*, *A. lamarcki*, *L. cucullata*, and *P. astreoides*. These coral populations (497 colonies in July 1977) were monitored each year until 1980.

Although the quadrats at Pinnacle 1 and Rio Bueno were deliberately placed in areas of high coral density to maximize numbers of monitored colonies, densities within the quadrats in 1977 were only slightly greater than average on these reefs (44 vs. 37% cover by platey corals at Pinnacle 1, 30 vs. 26% at Rio Bueno, see Tables 1, 2). Furthermore, because of the well-known patchiness of coral reefs (e.g., Grassle 1973), these differences were not statistically significant (Mann-Whitney *U* test, *P* > .1).

Photography

The 24 1-m² quadrats at Pinnacle 1 and Rio Bueno were photographed annually in July from 1977 to 1980 using a Nikonos III camera with a 28-mm lens and underwater strobe. Every quadrat was divided by monofilament line into four ¼-m² sections. Using color slide film, each ¼ m² was centered in the viewfinder and photographed from a perpendicular distance of 1–1.5 m, giving minimal lateral distortion and a large depth of field. Several hundred additional photographs

taken each year at different scales were useful for orientation, identification of particular corals, etc. All the species studied were easily identifiable in situ, with the exception of *A. lamarcki*, which we may have confused on occasion with the very similar *A. grahamae* Wells.

Each ¼-m² photograph was projected with a scale onto a screen, and all platey coral colonies were traced out, numbered, and digitized (a total of 1273 colonies over the 3 yr). We defined a colony as any physically separate patch of coral tissue growing independently of its neighbors (Connell 1973). Foliaceous corals often undergo fission caused by partial loss of tissue (Hughes and Jackson 1980), and undoubtedly many of the corals first monitored in 1977 were already represented by more than a single colony (i.e., several daughter colonies) because of fission prior to our study. Individual colony areas (to the nearest 0.5 cm²), as well as the combined areas of known daughter colonies (those formed by fission after 1977), were followed each year. All colonies were first assigned in 1977 to one of four size-classes (0–10 cm², 10–50 cm², 50–200 cm², and >200 cm²); these size-classes were chosen to encompass the range of colony areas in the different species, while maintaining adequate sample sizes. Then, amounts of growth and shrinkage in area of surviving colonies were followed each year.

In addition to net changes in areas of live corals, the area of tissue killed between censuses was recorded every year. The magnitude of tissue loss from each colony was scored from zero (escape from injury) through various amounts of injury (partial colony mortality or polyp death), to 100% (whole-colony mortality). Note that our definition of injury is based on loss of tissue alive in year 1 that had died in year 2, and

TABLE 2. Abundance of corals at Pinnacle 1 and Rio Bueno. Counts were made from annual photographs taken within 12 1-m² quadrats at each site (all depths combined at Rio Bueno) from 1977 to 1980.

A. Number of colonies (annual mean ± sd)				
Pinnacle 1	<i>A. agaricites</i> 175.0 ± 10.7	<i>A. lamarcki</i> 34.3 ± 1.5	<i>L. cucullata</i> 111.3 ± 7.5	<i>M. annularis</i> 88.8 ± 2.5
Rio Bueno	<i>A. agaricites</i> 239.5 ± 9.0	<i>A. lamarcki</i> 26.5 ± 2.4	<i>L. cucullata</i> 136.5 ± 15.8	<i>P. astreoides</i> 74.3 ± 2.2
B. Percentage cover of population in quadrats (annual mean ± sd)				
Pinnacle 1	<i>A. agaricites</i> 4.1 ± 0.2	<i>A. lamarcki</i> 12.9 ± 0.2	<i>L. cucullata</i> 7.9 ± 0.3	<i>M. annularis</i> 20.5 ± 0.3
Rio Bueno	<i>A. agaricites</i> 15.2 ± 0.6	<i>A. lamarcki</i> 5.4 ± 1.7	<i>L. cucullata</i> 2.9 ± 0.4	<i>P. astreoides</i> 4.7 ± 0.2

not on overall shrinkage of total colony area between photo-censuses. Partial mortality usually did result in a detected reduction of colony size, but corals often suffered moderate loss of tissue and still increased in area if net growth was fast. Therefore, use of simple changes in colony area would have underestimated injury rates, particularly for rapidly growing colonies.

An annual censusing regime like ours is apparently adequate to detect all but the smallest injuries in these species. Bak and Van Es (1980) have shown that experimentally inflicted 5-cm² circular lesions on the surface of *A. agaricites* and *P. astreoides* always remained unclosed even after 5 mo, while 50–90% of 1-cm² lesions (depending on species and depth) failed to regenerate over the same time period. Similar results were found for *M. annularis* (Bak et al. 1977).

When a coral suffered much partial mortality, surviving portions were often separated by the destruction of intervening tissue. Unlike branching corals, daughter colonies of foliaceous corals are not dispersed by waves, and all fission products stay in the same place (see review by Highsmith 1982).

In contrast to fission, when two physically separate colonies of the same species grew laterally and came into contact, their skeleton and tissues often fused. Typically, a postfusion colony looked at first like a figure eight, but soon grew into the shape of a normal colony. Grafting experiments indicated that fusion occurred only between coral fragments of the same original colony (T. P. Hughes, *personal observation*), as has been found for many other corals (cf. Potts 1976, Hildeman et al. 1977, Bothwell 1981, Neigel and Avise 1983).

Lateral growth rates

The amount of annual lateral growth in corals is small compared to the resolution of our photographs, and often occurs at slightly different angles to the camera. These two sources of error make it impossible to measure linear growth rates of individual colonies reliably from annual photographs, even for plating corals. Accordingly, we estimated lateral coral growth using in situ measurements and Alizarin-Red vital stain.

In situ measurements.—Before any measurements were taken we cemented with marine epoxy ≈1000 4-cm galvanized nails onto bare reef substratum close to the edges of corals within the quadrats, for use as fixed reference points. Radial growth of colonies was measured to the nearest millimetre with a plastic tape-measure held normal from the coral edge to the nearest nail. In a few cases growth was measured along more than a single edge (usually because there happened to be numerous nails around the coral), and these were averaged to give a mean growth rate per colony. Repeated measurements annually for 3 yr yielded growth rates. Because of colony mortality and loss or overgrowth of nails, the total number of measurements (1169 in 1977) declined by ≈50% each year.

Alizarin staining.—Alizarin staining was used at Rio Bueno as a check on the accuracy of the growth measurements, and at the Deep Mounds. Alizarin-Red is a vital dye which is incorporated into the underlying skeleton of a coral wherever it is covered by live tissue, giving a brightly colored mark showing the colony size at the time of staining (Barnes 1973). Skeleton deposited external to the stain is new growth. At Rio Bueno 83 colonies were successfully stained in situ by incubating each colony for a day or two in a clear polythene bag containing an Alizarin solution (20 mg/L). At the Deep Mounds, staining bags, 2000–4000 L in size, were weighted down with chain to enclose entire mounds and left in place for 3 or 4 d.

Colonies were collected 12 or 24 mo after staining, and if necessary were sectioned with a rock saw to reveal the stain. In very thin plates (especially deep-water specimens of *Agaricia* spp.) the alizarin could often be seen in whole skeletons. Lateral growth was measured as the distance between the stain and the new periphery of the plate (mean of 5–10 measurements per colony, depending on size) using a dissecting microscope fitted with a micrometer.

NUMBERS AND SIZES OF COLONIES

The numbers of colonies of each species are presented in Table 2, averaged over the four annual sets of observations. *Agaricia agaricites* followed by *L. cu-*

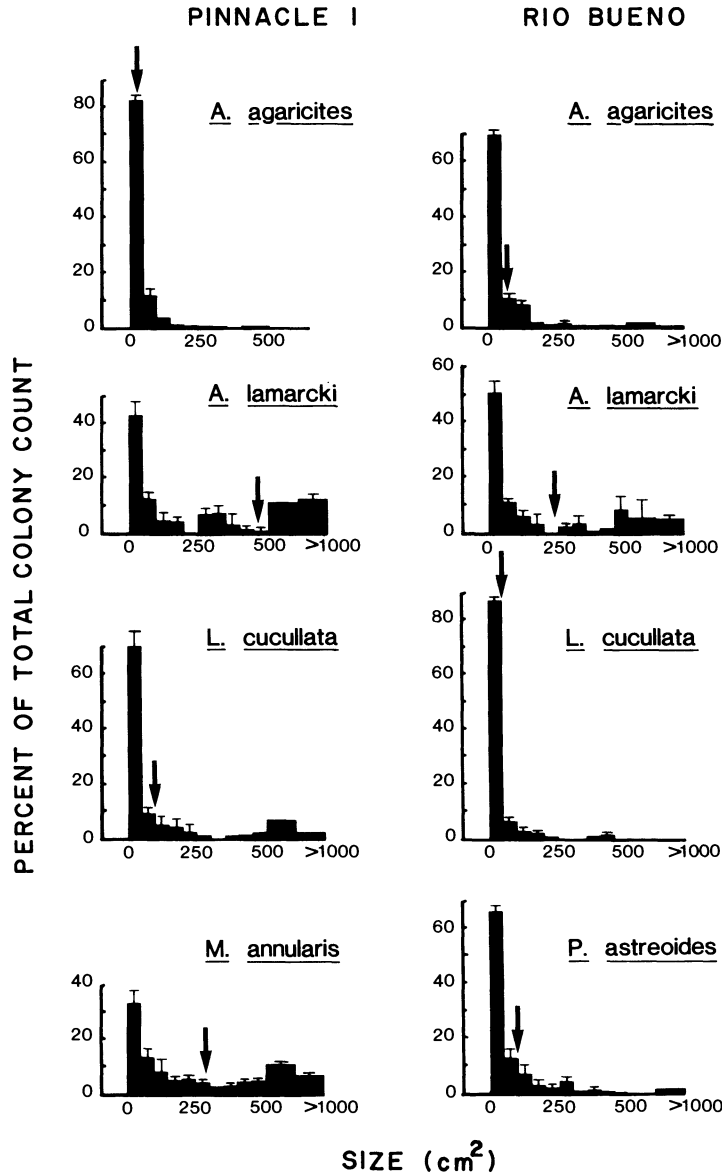


FIG. 1. Size-frequency distributions of colonies of *Agaricia agaricites*, *A. lamarcki*, *Leptoseris cucullata*, and *Montastrea annularis* at Pinnacle 1, and of *A. agaricites*, *A. lamarcki*, *L. cucullata*, and *Porites astreoides* at Rio Bueno. Data are for populations enclosed within 12 1-m² quadrats at each site. Because larger colonies were rarer, size-class increments were increased for the bigger corals (>500 cm²) from 50 to 500 cm². The two bars above 500 cm² on each graph thus represent sizes from 500–1000 cm², and >1000 cm², respectively, if present. Number of colonies per size-class is given as the mean annual (1977–1980) percentage of total colony count $\pm$ standard deviation. Sample sizes ranged from an annual mean of 27 to 240 colonies (see Table 2). Mean colony size is shown by an arrow in each graph.

cucullata have the most colonies at both sites, and *A. lamarcki* the least. Thus, the relative abundance of colonies for each species was identical at Pinnacle 1 and Rio Bueno, except for the virtual replacement of *P. astreoides* by foliaceous *M. annularis* at the deeper site (Table 2). At Pinnacle 1 relative abundance of the five species, measured in area covered, is generally opposite to that expected from numbers of colonies, and there is no apparent relationship between these two measures of abundance at Rio Bueno.

This disparity between colony counts and amount of cover indicates major differences between species in the relative numbers of small vs. large corals. This is seen clearly in a plot of the size-frequency distribution of colonies in each of the five coral species, based on 50-cm² increments (Fig. 1). The smallest corals (0–50 cm²) were the most common for all species at both sites. At Pinnacle 1, the percentage of the population smaller than 50 cm² ranged from 33–84% of the total colony counts, while at Rio Bueno, between 51–87%

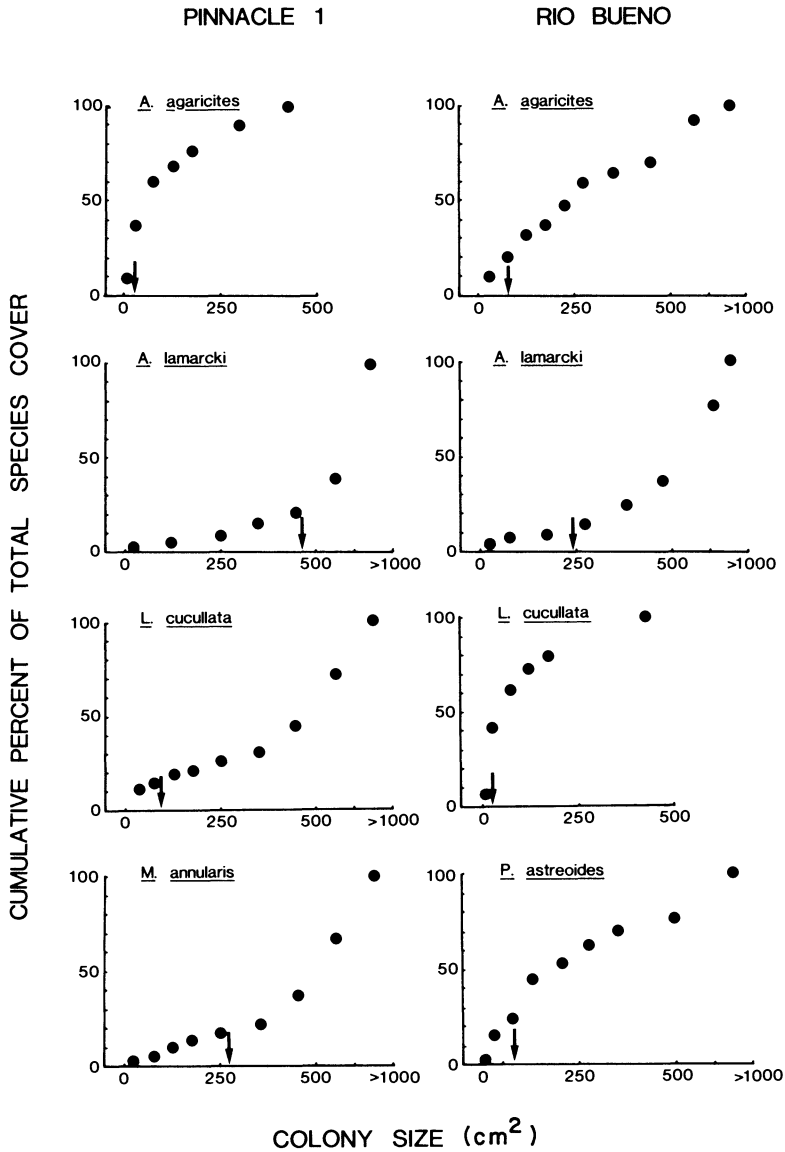


FIG. 2. Cumulative percentage of total species cover vs. colony size. The total absolute cover is given in Table 2. Mean colony size is shown by an arrow in each graph.

of colonies were in this smallest size-class. In general, larger corals became steadily rarer (so that the biggest colonies had to be plotted in 500-cm² rather than 50-cm² categories), although by far the steepest drop in abundance occurred at a very small size (Fig. 1). Nevertheless, there were considerably more large colonies of *A. lamarcki* and *M. annularis* than of the other three species. Mean colony area for both *M. annularis* and *A. lamarcki* was >200 cm², whereas for the others it was always <100 cm² (arrows in Fig. 1).

Despite the overwhelming number of small corals, they constitute very little of the total cover of any species (Fig. 2). The 0–10 cm² size-class makes up only from 1–8% of the total species area, even though these small colonies constitute up to half of the total colony counts.

Colonies smaller than the mean size account for only 10–25% of the areas of the various populations. In contrast, colonies >200 cm² make up between 21–96% of the various species areas, even though numerically they account for only 1–49% of the total counts.

The population size and size-frequency distributions of each species were remarkably constant over the four annual sets of observation (low standard deviations in Table 2 and Fig. 1), implying that during this time there was a balance in the processes which added to and took away from the number of colonies in any given size-class. Despite this apparent equilibrium, the number of colonies surviving through successive age-classes cannot be estimated from size-frequency data, because coral size and age are often poorly correlated

(Hughes and Jackson 1980). The size-frequency distribution of a coral species is not equivalent to a population structure in the normal usage of the term because small colonies can be old, due to partial mortality, fission, and fusion. The resemblance of Fig. 1 to a type III survivorship curve is both fortuitous and misleading; it cannot be used to make a time-specific (vertical) life-table.

DYNAMICS OF POPULATIONS

Here we consider the processes that affect the numbers of polyps and colonies of corals at Pinnacle 1 and Rio Bueno. These broadly include mortality, fission and fusion, and larval recruitment. We present the data in terms of events per unit time per individual colony rather than per unit area because (1) this is conventional for demographic analysis, and (2) the latter method would not allow easy comparisons between species or sites with different amounts of coral cover.

Mortality

There are three scales of mortality in colonial corals which may involve: (1) some but not all polyps in a colony (partial colony mortality), (2) entire colonies, and (3) entire genets. All of these reduce the amount of live tissue cover, but numbers of colonies and genets vary with the different forms of mortality in a more complex manner. By definition, partial mortality does not decrease number of colonies, but it may increase their number by promoting colony fission. Whole-colony mortality decreases the number of colonies in a population, but not necessarily the number of genets. Genet death occurs only when all the polyps and colonies derived from the same zygote are killed. Since fission almost certainly occurred repeatedly before our observations began, we can only identify as genets those colonies that settled after 1977. Therefore, we will focus primarily on the first two levels of mortality.

Partial and whole-colony mortality.—The mean annual frequencies of different degrees of colony mortality, ranging from no injury to complete colony death, are shown in Fig. 3. Of the 883 colonies monitored from 1977, 709 were injured or killed by 1980 (70% at Pinnacle 1, and 88% at Rio Bueno).

Patterns of partial and whole-colony mortality were strikingly related to colony size. The annual frequency of colony escape from injury was always highest in the smallest size-class (0–10 cm² in the first year of each set of observations). As colonies got larger, the probability of escape steadily decreased. For example, at Pinnacle 1, the frequency of colonies of each species that completely escaped injury each year was 64–74% for the smallest corals, 45–67% for size-class II, 17–57% for size-class III, and from 11–50% for the largest colonies. A similar trend was evident at Rio Bueno, although frequency of escape was generally lower (Fig. 3).

Frequency of whole-colony mortality was also higher

among small corals, decreasing steadily as colonies got larger. At Pinnacle 1, the frequency of colonies of each species killed outright each year was 14–27% for the smallest corals, and 0–18% for size-class II. No coral larger than 50 cm² (size-classes III and IV) was killed outright within 1 yr. At Rio Bueno, whole-colony mortality rates were higher, but were similarly size dependent.

Since the largest colonies had the lowest frequencies of escape and of whole-colony mortality, they obviously had the highest rate of injury. However, the relative amount of tissue lost from each bigger colony was generally less than for smaller colonies. Depending on the species, 0–23% of the corals injured in the largest size-class lost more than one-third of their tissue in any year. In comparison, up to 59% of injured 0–10 cm² corals, 56% of those 10–50 cm², and 45% of the 50–200 cm² size-class lost more than one-third of their area every year (Fig. 3).

Survivorship curves.—Many of the colonies that died by 1980 had earlier been observed undergoing fission into two or more daughter colonies, and often some daughter colonies survived when others (presumably of the same genotype) were killed. Thus the yearly loss of colonies (Fig. 3) will overestimate the loss of original 1977 colonies (Fig. 4). At Pinnacle 1, 21% of the original corals alive in 1977 died by 1980 (80 colonies), while at Rio Bueno 47% were killed (234 colonies). From 1977 to 1980, mortality rates for colonies of the different species ranged five-fold from 6 to 34% at Pinnacle 1, and two-fold from 38 to 63% at Rio Bueno.

In general, the smallest corals (0–10 cm² in 1977) had the highest mortality losses, which were especially heavy at Rio Bueno (Fig. 4). The depletion of cohort I over the 3 yr showed a range among the species (omitting populations with <10 individuals in cohort I initially) of 27–45% at Pinnacle 1, and 48–69% at Rio Bueno. Mortality in cohort II was slightly reduced, showing a range among species (for $n > 10$ colonies in 1977) of 23–42% loss at Pinnacle 1, and 37–58% at Rio Bueno. Colonies in cohorts III and IV, in contrast, showed very slight losses, although sample sizes were generally quite small (depending on species, ranging from 13 to 73 colonies for both 1977 size-classes combined). Combining data for all species, the mortality rate for colonies originally in cohort III was 12% at Pinnacle 1, compared to a 29% loss at Rio Bueno. For cohort IV, the loss at the two sites was 0 and 18% respectively.

Partial mortality killed more tissue than did the mortality of entire colonies (Fig. 5). The importance of partial colony mortality is seen most clearly in deep-water *A. lamarcki* and *M. annularis*, where virtually all of the tissue lost was through injuries (Fig. 5). In contrast, *Agaricia agaricites* and *L. cucullata* at Rio Bueno (but not at Pinnacle 1) lost a good deal of their tissue through death of whole colonies, probably because of their relatively limited size-range, dominated

PINNACLE I

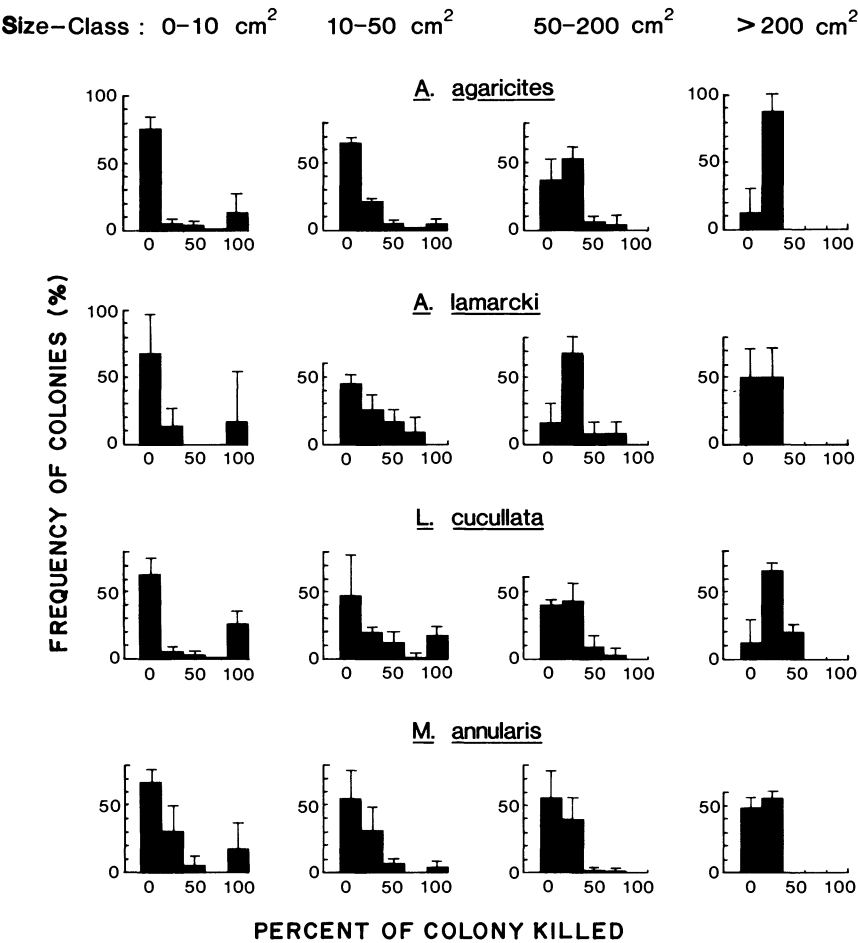


FIG. 3. Mean annual frequency distribution of the percentage of colony mortality for four size-classes. Size-class dimensions are as follows: I (0–10 cm²), II (10–50 cm²), III (50–200 cm²), and IV (>200 cm²). Error bars show one standard deviation around mean annual frequencies. Percentage colony mortality was scored from zero (escape from injury) through various amounts of tissue loss (1–33%, 33–66%, and 66–99%), to 100% (whole-colony mortality). Colonies were assigned to the various size-classes on the basis of their area in the first year of each of the pairs of quadrat photographs (1977/1978, 1978/1979, 1979/1980). The mean annual sample sizes (number of colonies) for the smallest to the largest size-classes were as follows: at Pinnacle I, *A. agaricites* (76, 72, 24, 4), *A. lamarcki* (6, 7, 4, 15), *L. cucullata* (39, 41, 18, 11), and *M. annularis* (8, 22, 19, 36); at Rio Bueno, *A. agaricites* (101, 65, 49, 24), *A. lamarcki* (9, 10, 6, 11), *L. cucullata* (70, 49, 18, 2), and *P. astreoides* (24, 26, 18, 7 colonies).

by short-lived, small colonies (Fig. 1). However, even for these two populations, partial and whole mortality were almost equally important.

Temporal variability and storms.—Loss of coral tissue varied from one year to the next (Fig. 5), and was particularly heavy in year II (1978/1979) because of a severe storm in March 1979 (Tunnicliffe 1980, Hughes 1984). In year I before the storm (1977/1978), the amount of coral destroyed varied from 4–14% of the species cover at Pinnacle 1, and between 5 and 43% at Rio Bueno. In year II, tissue losses increased greatly in many species, especially at Rio Bueno, where the total amount of coral destroyed more than doubled. In year III (1979/1980), partial and whole-colony mor-

talities generally declined to year I levels (Fig. 5). However, deepwater *A. lamarcki* and *M. annularis* did not show these trends, and neither did the *P. astreoides* population at Rio Bueno.

Also in year II, there was a marked increase in amount of cover lost from whole-colony mortality (Fig. 5), caused by increased loss of large corals, many of which were apparently torn off the reef in rough seas. Interestingly, Hurricane Allen, which struck Jamaica in August 1980 after our annual census, had almost no effect on these populations, probably because vulnerable corals had already been selected out the previous year (Hughes 1984). Destruction from the 1979 storm may have been particularly severe because of the length of

RIO BUENO

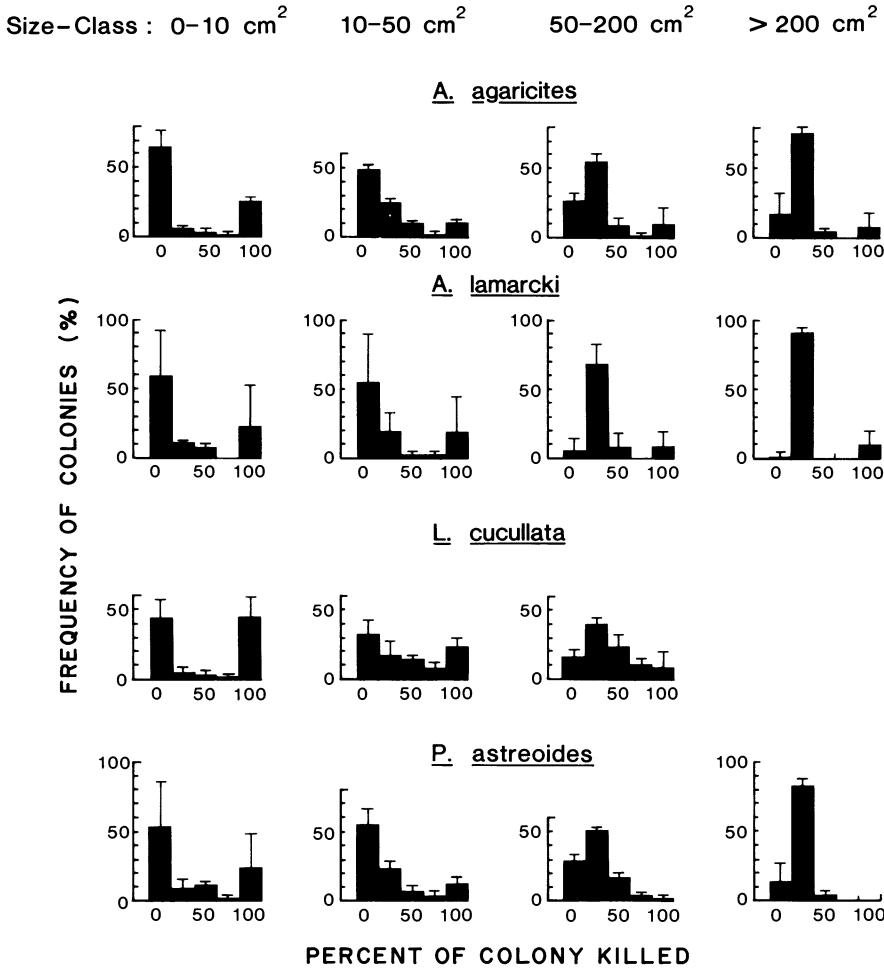


FIG. 3. Continued.

time since the preceding major disturbance; destructive winter storms apparently occur only every 20 yr or so, and the most recent severe hurricane before this storm was in 1917 (Tunncliffe 1980).

Tissue turnover rates.—Although there has been much debate concerning the stability of coral reefs, there are no published measurements on the rates of turnover of coral tissue. In the present study, the time required for each population to lose the amount of tissue equivalent to its cover in 1977, based on average rates for 1977–1980, varied from 5–32 yr at Pinnacle 1, and between 3–6 yr at Rio Bueno (Table 3). Even if this calculation were based only on the year with the lowest amount of injury and mortality (usually 1979/1980), the length of time required for total tissue turnover is relatively short compared to the potential lifespan of individual colonies of these species.

The amount of tissue destroyed at Pinnacle 1 from 1977 to 1980 varied six-fold depending on species, up

to a maximum of 56% of the original cover, while at Rio Bueno there was a two-fold range with a maximum loss of close to 100% (Table 3). Despite this heavy damage, many of the populations showed a net gain in percentage cover during the 3 yr, because growth of surviving colonies (and, to a small extent, the arrival of new sexual recruits) exceeded tissue losses. Therefore, the rate of tissue turnover does not indicate whether a population is expanding or declining.

Historical effects.—Partial and whole-colony mortality were very patchy in space as well as time. Often colonies were killed or badly injured while their neighbors were unharmed. Some of this variability was due to size or species differences (Figs. 3–5); some may have been caused by microenvironmental and genetic variation, or by chance. Analysis of the photographs showed that past experiences of corals also affected which were damaged or escaped (Hughes 1984).

All corals surviving year I or year II were classified

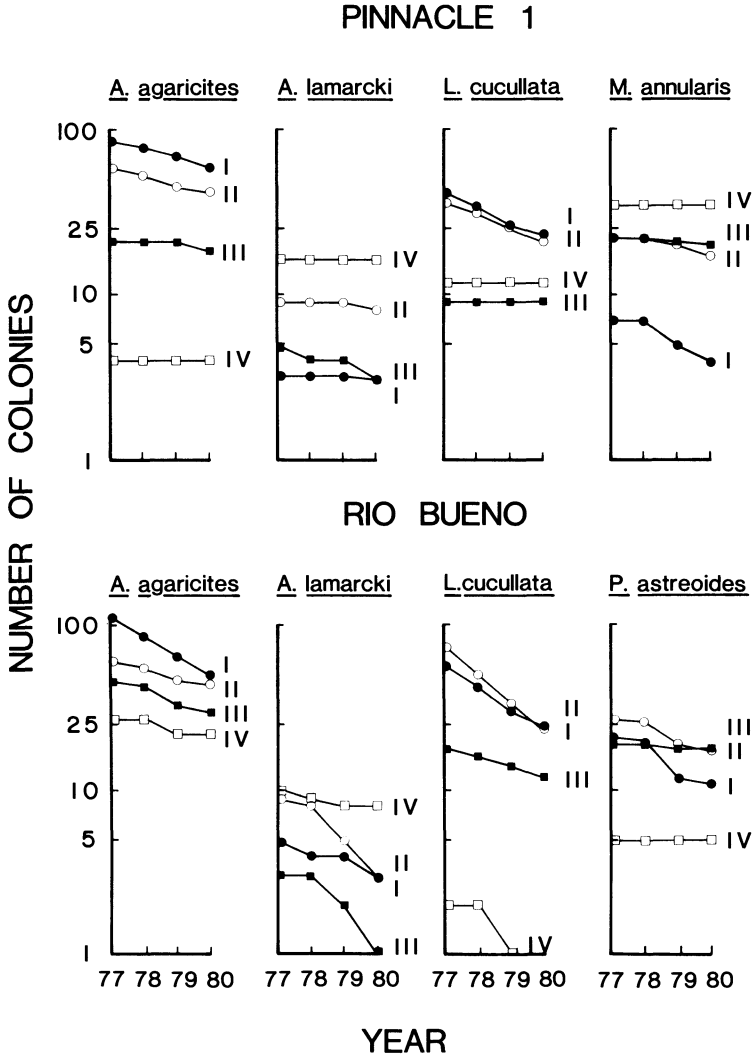


FIG. 4. Survivorship curves for whole colonies, as a function of their size in 1977. Size-classes are defined in Fig. 3.

as to whether or not they were injured during that year. Their fates were then recorded for the next year, II or III, respectively, and this new outcome compared to that of the year before. The resultant data formed the basis for two 2×3 contingency tables, whose row classifications were injury or escape in year I, and whose column classifications were injury, escape, or whole-colony mortality in year II. If the chi-square values were significant for both sets of data (year I equals 1978 or 1979, respectively), data were combined into a single table for each population.

With only one exception, all species at both sites consistently showed a highly significant difference in the fate of colonies depending on whether they were injured or not the year before (Fig. 6). Specifically, surviving corals were about twice as likely either to escape again or to be injured repeatedly, rather than have an escape followed by an injury or vice versa.

The one exception was *A. lamarcki* at Pinnacle 1, which showed no historical effect ($\chi^2 = 0.14$, $P = .7$), despite a highly significant outcome at Rio Bueno ($\chi^2 = 16.26$, $P < .001$). Otherwise, the frequency of outcomes in year II for corals which had escaped injury in year I was always: escaped again > injured > killed. The ranking for corals that had been injured in year I was always injured again > escaped or killed (Fig. 6).

To test whether such historical effects last for periods longer than a single year, the same procedure was repeated to draw up contingency tables of colony fate in year III as a function of experiences in year I. Only two of the eight populations, *A. agaricites* and *P. astreoides* at Rio Bueno, showed a significant result, in the same direction as in Fig. 6 ($\chi^2 = 10.45$ and 8.70 , $P < .01$ in both cases). The overall decrease in the amplitude of historical effects from 1 to 2 yr following the event may be due to the recovery of injured colonies

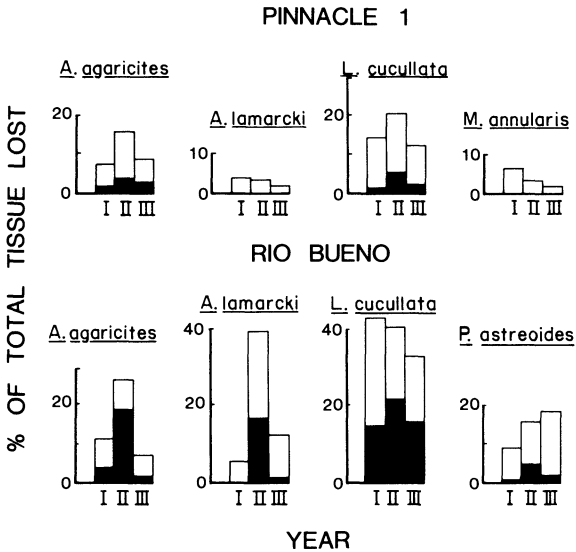


FIG. 5. The amount of tissue lost each year through partial and whole-colony mortality (year I = 1977/1978, year II = 1978/1979, year III = 1979/1980). The area of tissue destroyed was standardized by dividing by the total area of live coral for each population in the 1st yr of each 3-yr set of photographs. Absolute amounts of tissue loss are given in Table 3.

with time, so that future demographic differences between them and injury-free colonies become unpredictable (Hughes 1984).

Causes of mortality.—Less than 6% of coral injuries or deaths were due to overtopping (defined as the death of some or all of a shaded coral caused when the perimeter of another colony overgrew it in the horizontal plane). We never observed extra-coelenteric fighting, but we did notice an unusual form of coral-coral “interaction” (accounting for 4% of injured or killed platey

colonies) which sometimes resulted in displacement. Loose fragments of the branching corals *Acropora cervicornis* and *Porites furcata* were occasionally washed onto plating corals during storms, shattering the delicate, foliaceous skeletons. Many of the larger fragments of the branching species survived and became established in place of the plating corals.

A further 727 colonies (90% of the total affected corals) were damaged or killed by processes other than the coral-coral interactions described above. It appears likely that these corals interact far more frequently with other reef organisms than among themselves (see also Bak et al. 1976, Ohlhurst 1980, Suchanek et al. 1983). Most of the dead coral surfaces were covered by sediment or new algal growth, but we cannot tell from annual photographs if sedimentation or competition with short-lived plants was the source of coral damage, or if they were preceded by events such as predation or other forms of disturbance. Regardless, it would appear that the most thoroughly studied agents of coral mortality are relatively unimportant at our sites.

Fission and fusion

A total of 123 fission events (involving 89 original colonies) were counted at Pinnacle 1 and Rio Bueno between 1977 and 1980 (Table 4). An average of 2.54 daughter colonies were formed at each event, producing initially 189 additional colonies, of which 128 survived until 1980 (Table 4).

Just over half of the fissions took place in the storm year II. The range between species in mean annual frequency of colony fission was almost identical at the two sites, varying from 0.02 to 0.10. The higher probability of fission for *L. cucullata* at Pinnacle 1 (Table 4) probably reflects the presence of nearly six times more size-class IV colonies (>200 cm²) at this site compared to Rio Bueno. For all species at Pinnacle 1,

TABLE 3. Estimates of the number of years required for 100% destruction of the original (1977) area of live tissue within the 12 1-m² quadrats at each site. Average turnover time was calculated using the mean annual amount of tissue lost for the three years shown in Fig. 5. Maximal turnover time was calculated using data for the year with the least amount of tissue destruction for each species.

A. Pinnacle 1	Species			
	<i>A. agaricites</i>	<i>A. lamarcki</i>	<i>L. cucullata</i>	<i>M. annularis</i>
Population area in 1977 (10 ³ cm ²)	4.64	15.31	9.29	24.74
Area of tissue destroyed 1977–1980 (10 ³ cm ²)	1.54	1.43	5.18	2.89
Percent destroyed	33.2	9.3	55.8	11.7
Average turnover time (yr)	9	32	5	26
Maximal turnover time (yr)	14	47	8	52

B. Rio Bueno	Species			
	<i>A. agaricites</i>	<i>A. lamarcki</i>	<i>L. cucullata</i>	<i>P. astreoides</i>
Population area in 1977 (10 ³ cm ²)	18.07	7.97	4.10	5.34
Area of tissue destroyed 1977–1980 (10 ³ cm ²)	8.55	4.54	4.20	2.54
Percent destroyed	47.3	57.0	102.5*	47.5
Average turnover time (yr)	6	5	3	6
Maximal turnover time (yr)	15	18	4	11

* Because of new larval recruitment, growth, and regeneration of damaged tissue, the area of *L. cucullata* tissue destroyed from 1977–1980 exceeded the original amount present.

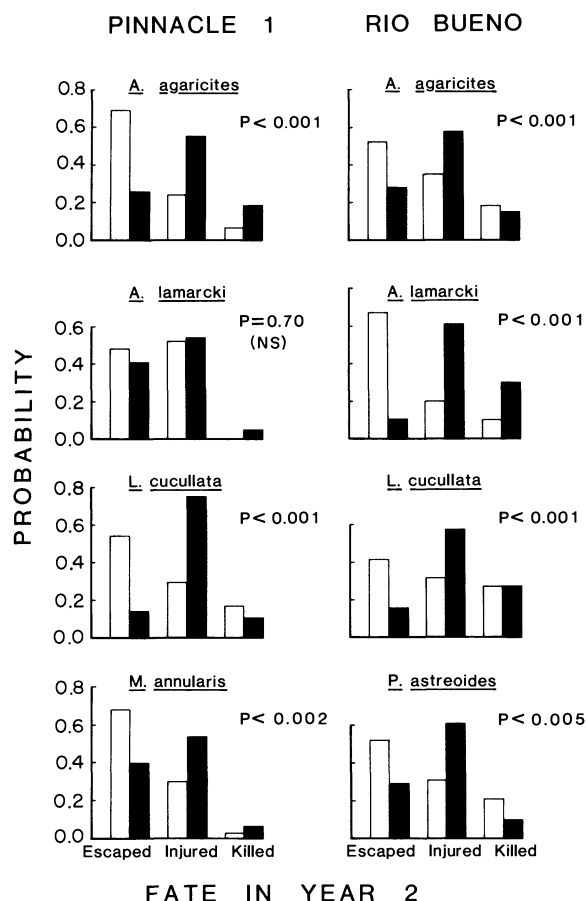


FIG. 6. The fate of colonies in year II (1978/1979 or 1979/1980) as a function of whether they were injured (■) or not (□) in year I (1977/1978 or 1978/1979). Colonies in year II were scored as having escaped, been injured, or been killed outright. The numbers of colonies which had escaped or were injured (the latter in parentheses) in year I were as follows: at Pinnacle 1, *A. agaricites*: 217 (89), *A. lamarcki*: 25 (39), *L. cucullata*: 95 (56), *M. annularis*: 81 (83), and at Rio Bueno, *A. agaricites*: 199 (170), *A. lamarcki*: 15 (31), *L. cucullata*: 102 (76), *P. astreoides*: 61 (59).

colonies in this largest size-class were almost 20 times more likely to undergo fission than the smallest corals, and 35 times more likely at Rio Bueno (Fig. 7). The difference between sites reflects higher rates and amounts of injury at Rio Bueno (Figs. 3–5). The size-frequency distribution of colonies before and after fission is shown for each species in Table 4. Naturally, colonies are on average much smaller after fission than before.

A total of 77 corals fused with a neighboring colony during the 3 yr (Table 5). The majority of fusions involved only two colonies (34 out of 37 events), although rarely three colonies coalesced in a single year. Fully 92% of the fused colonies survived to the end of the study, in contrast to the sharp decline in the number of colonies following fission (Table 4). When data from the various species were combined, there was no con-

sistent relationship between colony size and frequency of fusion. There was also no correlation between rates of fission and of fusion (Spearman rank correlation 0.4 at Pinnacle 1, and 0.6 at Rio Bueno, $P > .1$ in each case), probably due in part to differences between species in the survival rates of daughter colonies (Table 4).

Larval recruitment

We detected newly settled corals after they reached a diameter of 5–10 mm, corresponding normally to an age of <1 yr (Vaughan 1911, Bak and Engel 1979, Rylaarsdam 1983). Larval recruits are easily distinguishable from new daughter colonies because (1) they are smaller than most postfission colonies, (2) larvae often settle on coral rock made by a different species, while daughter colonies are connected to their neighbors by a common skeleton, and (3) daughter colonies retain part of the original colony outline for at least a year after fission has occurred, and are always the same color.

A total of 201 larval recruits were observed. Recruitment was much higher at Rio Bueno than at Pinnacle 1 (4.2 vs. 1.2 recruits·m²·yr). Settlement rates were quite constant during the 3 yr at Rio Bueno, but not at Pinnacle 1, where 73% of the larval recruitment occurred in year I.

The species differed widely in their recruitment rates and subsequent survival of young colonies (Table 6). Not unexpectedly, species with the highest rates of settlement also had size-frequency distributions dominated by small colonies throughout the study (Fig. 1). At both sites, *L. cucullata* and *A. agaricites* had the greatest recruitment; *A. lamarcki* and *M. annularis* showed quite low recruitment; *P. astreoides* was intermediate (Table 6).

Population stability

We can now summarize the underlying dynamics of the superficially static populations over the 3 yr (Table 7). There were 883 colonies of the five species at Pinnacle 1 and Rio Bueno in 1977, of which just over one-third died. Additional colonies were added in roughly equal numbers through partial mortality (fission) and larval recruitment. Ironically, although fission is an important source of tissue loss, the number of new colonies generated by partial mortality offset 40% of the death of whole colonies. Similarly, fusion reduced colony counts by an amount equivalent to 13% of the recorded losses from whole-colony mortality. The net result of all this turmoil was only a 9% decrease in numbers of colonies, despite the unusually severe winter storm of 1979.

There was somewhat more variability in colony numbers among the five species. Colony counts during the 3 yr deviated from 1977 values by <10% for *A. agaricites*, *A. lamarcki*, and *M. annularis* at Pinnacle 1, and for *A. agaricites* and *P. astreoides* at Rio Bueno (Fig. 8). However, numbers of colonies increased by

TABLE 4. The number of parent (pre-fission) and daughter (post-fission) colonies classified by site, species, and size-class, for all years combined. The mean annual probability of fission was calculated by dividing the yearly number of parent colonies by the total population size (number of colonies).

	Species							
	<i>A. agaricites</i>		<i>A. lamarcki</i>		<i>L. cucullata</i>		<i>M. annularis</i>	
	Before	After	Before	After	Before	After	Before	After
A. Pinnacle 1								
Area of colony before and after fission								
0–10 cm ²	1	8	0	3	1	24	0	6
10–50 cm ²	3	11	1	0	4	31	2	5
50–200 cm ²	6	1	0	1	14	22	2	1
>200 cm ²	0	0	1	1	11	9	7	10
Total no. of colonies	10	20	2	5	30	86	11	22
No. of survivors in 1980	17		3		68		21	
% of 1980 population	10.4		9.9		59.6		24.4	
Mean annual probability of fission	0.020		0.020		0.100		0.043	
B. Rio Bueno								
	Species							
	<i>A. agaricites</i>		<i>A. lamarcki</i>		<i>L. cucullata</i>		<i>P. astreoides</i>	
	Before	After	Before	After	Before	After	Before	After
Area of colony before and after fission								
0–10 cm ²	2	29	0	3	1	33	0	17
10–50 cm ²	12	37	0	2	5	16	3	8
50–200 cm ²	12	12	1	8	8	5	4	1
>200 cm ²	12	12	7	4	1	0	2	2
Total no. of colonies	38	80	8	17	15	54	9	28
No. of survivors in 1980	74		14		37		17	
% of 1980 population	31.8		60.9		32.2		38.4	
Mean annual probability of fission	0.051		0.099		0.033		0.041	

just less than 20% in *L. cucullata* at Pinnacle 1, and decreased by 20 and 25%, respectively, in *A. lamarcki* and *L. cucullata* at Rio Bueno (Fig. 8).

CORAL GROWTH

As well as numbers of colonies, population size structures also varied little throughout the 3 yr (low variance in Fig. 1). This implies a balance in processes that changed sizes of individual colonies, so that colonies added to size-classes made up for losses. In the following sections we describe changes in coral size due to growth and shrinkage. We calculated both lateral and areal growth because these are not equivalent, e.g., a coral can extend its edge in one direction while partial mortality of another part of the colony simultaneously reduces its area.

Linear growth

Virtually all the foliaceous corals in the quadrats at –10, –20, and –35 m were measured repeatedly, regardless of colony condition, size, or ease of access. Consequently, we have avoided as far as possible any bias inherent in selection of corals for growth analysis. A large proportion of corals was injured between the annual sets of measurements (Fig. 3), but these corals are included in our analysis if net growth along the measured coral radius was >0. Negative growth is treated in terms of changes in colony area in the next section.

The most striking result was the variability of the data (Fig. 9). In general, the modal extension rate of colonies was slower than the mean at each depth, reflecting the skewed distribution of growth rates within each population. Typically, corals most frequently grew <5 mm per year, with fewer and fewer individuals in

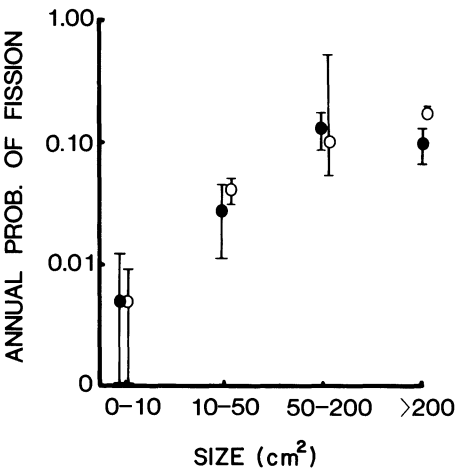


FIG. 7. The annual probability of fission (mean ± SD) vs. colony size for all species combined (from Table 4). The total number of colonies that underwent fission in the smallest to largest size-classes was 2, 10, 22, and 19 at ● Pinnacle 1, and 3, 20, 25 and 22 at ○ Rio Bueno.

TABLE 5. The number of pre- and post-fusion colonies categorized by site, species, and size-class, for all years combined. The mean annual probability of fusion was calculated by dividing the yearly number of pre-fusion colonies by the total population size (number of colonies).

	Species							
	<i>A. agaricites</i>		<i>A. lamarcki</i>		<i>L. cucullata</i>		<i>M. annularis</i>	
	Before	After	Before	After	Before	After	Before	After
A. Pinnacle 1								
Area of colony before and after fusion								
0–10 cm ²	1	0	0	0	0	0	0	0
10–50 cm ²	2	1	0	0	1	0	6	0
50–200 cm ²	2	1	1	0	1	1	2	3
>200 cm ²	1	1	3	2	0	0	2	2
Total no. of colonies	6	3	4	2	2	1	10	5
No. of survivors in 1980	3		2		1		5	
% of 1980 population	1.8		6.1		0.9		5.8	
Mean annual probability of fusion	0.011		0.039		0.006		0.037	
	Species							
	<i>A. agaricites</i>		<i>A. lamarcki</i>		<i>L. cucullata</i>		<i>P. astreoides</i>	
	Before	After	Before	After	Before	After	Before	After
B. Rio Bueno								
Area of colony before and after fusion								
0–10 cm ²	7	1	0	0	2	0	0	0
10–50 cm ²	21	4	0	0	4	1	2	0
50–200 cm ²	8	9	0	0	0	2	0	1
>200 cm ²	9	7	2	1	0	0	0	0
Total no. of colonies	45	21	2	1	6	3	2	1
No. of survivors in 1980	18		1		3		1	
% of 1980 population	7.7		4.3		2.6		1.4	
Mean annual probability of fusion	0.063		0.025		0.015		0.009	

progressively faster growth-classes. Because the fastest growing colonies were rare for all populations, the accuracy of the distributions of extension rate depends on sample size, and the number of coral colonies we have measured (778) is a comparatively small sample.

Reductions in growth rate with depth were slight, but nonetheless significant ($P < .05$ for each species; Kruskal-Wallis analysis of variance of ranks for species at three depths; Mann-Whitney U test for *P. astreoides*). However, because of the great variability, many of the faster growing colonies in deep water exceeded the extension rate of the majority of shallower specimens (Fig. 9). Thus, depth is a poor predictor of growth rate in these foliaceous coral populations.

Coral extension rates did not vary with initial colony size (Table 8). The relationship was tested by plotting radial growth vs. initial diameter of each colony. In no case was the slope of the relationship significantly different from zero (Spearman rank order correlations, $P > .10$ in all cases). Therefore, the smallest corals in our samples were capable of extending laterally as fast as the largest, and coral growth did not slow down or stop as colonies grew bigger. Individual corals grew until they were injured or killed, and there is apparently no genetically programmed maximum colony size. However, the lack of correlation between extension

TABLE 6. The number of new colonies originating as larval recruits observed in the photographs every year for each population. The total detected influx was 201 new colonies.

	Species			
	<i>A. agaricites</i>	<i>A. lamarcki</i>	<i>L. cucullata</i>	<i>M. annularis</i>
A. Pinnacle 1				
Year of settlement				
1977/1978	19	2	11	0
1978/1979	6	1	6	0
1979/1980	4	0	2	0
Total settlement	29	3	19	0
Survivors 1977–1980	25	3	16	0
% of 1980 population	15.6	9.1	13.7	0.0
	Species			
	<i>A. agaricites</i>	<i>A. lamarcki</i>	<i>L. cucullata</i>	<i>P. astreoides</i>
B. Rio Bueno				
Year of settlement				
1977/1978	25	1	27	2
1978/1979	23	0	22	6
1979/1980	21	0	20	3
Total settlement	69	1	69	11
Survivors 1977–1980	60	1	39	11
% of 1980 population	24.1	4.5	34.2	14.7

TABLE 7. Summary of events affecting the number of coral colonies over 3 yr for all five species at Rio Bueno and Pinnacle 1 combined.

Factor affecting population size	Change	Result
Original number of colonies (1977)		833
Number of colonies that died	-315	568
Daughter colonies formed by fission	+189	757
Number of daughter colonies that died	-61	696
Colonies lost by fusion	-40	656
Number of fusion colonies that died	-3	653
Number of larval recruits	+201	854
Number of recruits that died	-46	
Number of colonies (1980)		808

rate and size means that relative growth in area decreased as a coral got bigger, i.e., the length of time it took a colony to double its surface area increased steadily with larger coral size.

The ranking of mean extension rate among species was *P. astreoides* > *A. agaricites* > *A. lamarcki* > *M. annularis* (Fig. 9). We have no linear growth data for

L. cucullata, but from the growth measurements based on area it seems clear that this species grew the fastest of all (see below). However, many colonies of the slowest growing species grew faster than many colonies of the fastest species.

Growth in area of colonies

Changes in area of colonies (or combined areas of known daughter colonies formed after 1977) were strongly dependent on their initial size. Typically, small colonies (size-classes I and II in 1977) grew proportionately more, while larger colonies (originally in size-classes III and IV) grew less (Fig. 10). This was to be expected because small corals could extend laterally at the same rate as bigger colonies (Table 8) and would therefore tend to have proportionately much faster rates of relative areal growth. Furthermore, small colonies were less likely to lose area by partial mortality than were larger colonies (Fig. 3). Within each 1977 size-class, some colonies grew (or shrank) much faster than others, resulting in an increased variance around mean values through time. The nature of this within-cohort divergence is important from an evolutionary standpoint, and is examined more closely in the next section.

Differences between size-classes were most extreme in *A. agaricites* and *L. cucullata*, whose smallest colonies had a relative growth rate about six times faster

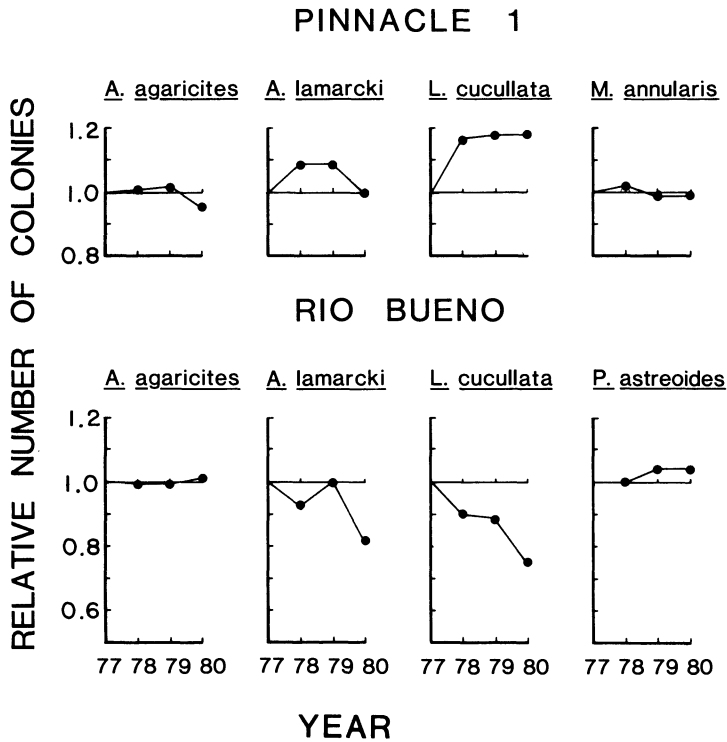


FIG. 8. Numbers of colonies (relative to 1977 counts) through time for each population. Counts include post-1977 larval recruits and asexually derived daughter colonies.

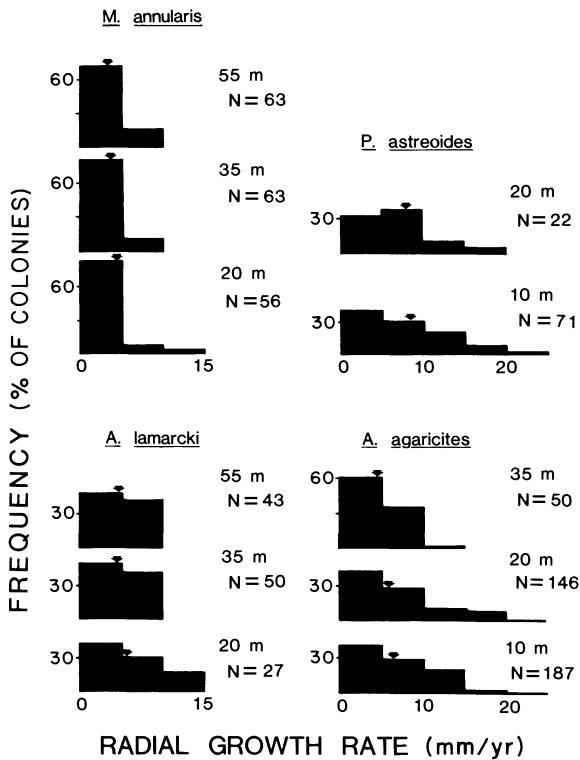


FIG. 9. Frequency distribution of annual radial growth rates at various depths for *A. agaricites*, *A. lamarcki*, *P. astreoides*, and *M. annularis*. Measurements were categorized into 5-mm growth-classes and the frequency distributions standardized by dividing by the sample size for each depth. The mean growth rate of each distribution is shown by the arrow (●) above the histogram. The 10-m and 20-m samples were from the quadrats at Rio Bueno, the 35-m sample from the quadrats at Pinnacle 1, and the 55-m sample from the Deep Mounds. The sample size (N) is the number of colonies measured in situ or after staining and collection. The total number of colonies measured over 3 yr was 778, including 240 stained colonies.

than the largest. Small colonies of these two species had the highest rates of whole-colony mortality, so that fast growth towards a larger and “safer” size would be of particular advantage. In contrast, *P. astreoides*, *A. lamarcki*, and *M. annularis* showed a less pronounced difference in growth rate as a function of colony size.

Population cover and regeneration

Remarkably, despite all of the tissue lost through colony mortality and injury, population area remained relatively stable during our investigations (Fig. 11). The disparity between absolute tissue loss (Table 3) and the net change in population area (Fig. 11) is due to the growth of surviving colonies, and to a trivial extent to new sexual recruits. Clearly, some populations are much better at recovering lost tissue than others (Table 9).

SIZE-CLASS TRANSITIONS

Combining the previous demographic and growth data we can now determine the probability that a particular colony (e.g., species *x*, size-class *y*, locality *z*) will survive and, if it does, the likelihood that it will grow into a larger size-class, stay the same size, or decrease into one or more smaller size-classes. With few exceptions, colonies from each size-class (in 1977) ended up scattered between three or four size-classes within 3 yr. For each “cohort,” the number of colonies per size-class in 1980 was standardized by dividing by the original 1977 sample size. These proportions represent transitional probabilities describing the frequency with which colonies moved from the original size-class to another (or stayed in the same size-class, if net change in area was small).

Transitional probabilities were highly dependent on original colony size (Fig. 12). Rates of partial colony mortality (Fig. 3) were lowest for the cohort of smallest colonies (1977 size-class I), which invariably showed significant growth towards larger categories (to size-class II and occasionally III), although in nearly every population many colonies were still in size-class I after 3 yr. In contrast, individuals in the cohort of largest colonies (size-class IV in 1977) showed little net growth because of partial mortality, and instead tended to “re-

TABLE 8. The relationship between colony size and linear (extension) rate. The Spearman rank correlation coefficient was calculated for amount of radial growth vs. initial diameter for each species. None of the correlations was significant.

Species	Depth (m)	No. colonies	Ranges		Correlation coefficient (r_s)
			Initial diameter (cm)	Radial growth rate (cm/yr)	
<i>A. agaricites</i>	10	187	0.6–78.3	0.0–2.4	–0.11
	20	146	0.8–63.4	0.0–2.3	–0.10
	35	50	0.8–41.1	0.0–1.5	+0.19
<i>A. lamarcki</i>	20	27	0.9–90.1	0.0–1.4	+0.25
	35	50	0.8–126.3	0.0–1.0	–0.18
	55	43	1.3–45.5	0.0–1.0	+0.21
<i>P. astreoides</i>	10	71	1.0–69.2	0.0–2.4	+0.16
	20	22	2.1–60.7	0.2–2.0	–0.34
<i>M. annularis</i>	20	56	15.2–57.2	0.0–1.5	+0.17
	35	63	2.3–183.7	0.0–1.0	–0.14
	55	63	1.1–40.0	0.0–0.9	–0.12

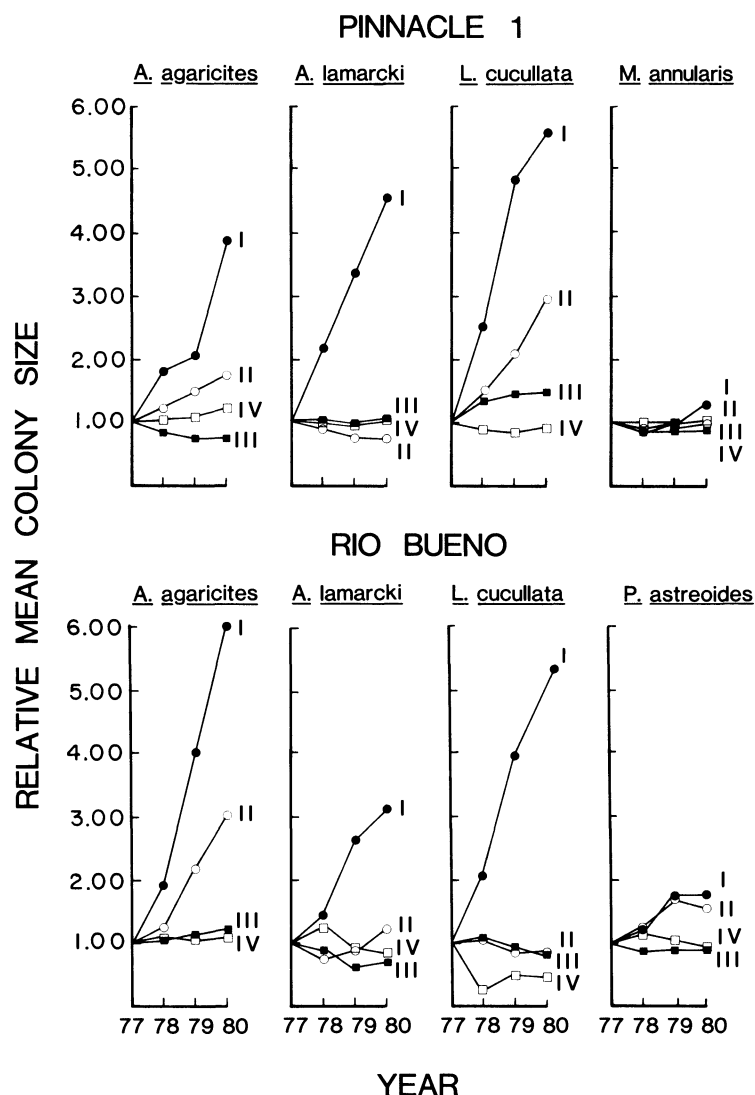


FIG. 10. Mean area of surviving individual colonies from 1977 through 1980 for corals in the four size-classes (defined in Fig. 3 legend) in 1977, relative to their size in 1977. Sample sizes are given in Fig. 1.

cruit" into smaller size-classes through shrinkage and fission. For all populations, the number of colonies alive in cohort IV actually increased over time (making the sum of transition probabilities > 1), because colonies formed by fission more than compensated for losses from mortality. The most extreme example was provided by *L. cucullata* at Pinnacle 1, where for every coral alive in cohort IV in 1977, there were more than three in 1980. Note that the formation of small colonies by fission did not greatly deplete the largest size-class, because large corals were usually injured by only a small amount during fission, with the largest daughter colony usually remaining in size-class IV (Table 4). Cohorts II and III were intermediate in character between the largest and smallest corals.

The transitional probabilities map out the diverse future of equal-sized colonies. For example, in 1977 there were 21 colonies of *A. agaricites* in size-class III

(50–200 cm²) at Pinnacle 1. By 1980, 3 of them had died and 4 of the surviving 18 had undergone fission, leaving 22 colonies. Three of the 1980 colonies were in size-class I, 9 were in size-class II, 10 were in size-class III, and none had grown into size-class IV (Fig. 12). Conversely, transitional probabilities can be used to trace backwards in time the origins of a group of (presently) like-sized corals.

LIFE HISTORIES OF FOLIACEOUS CORALS

Some important life-history features of the five species are summarized in Table 10. Each species was ranked from 1 to 5 with respect to larval recruitment, linear growth, proportion of small colonies, whole and partial mortality, and tissue turnover. Differences between species are striking. *Leptoseris cucullata* populations are dominated by very small colonies and show the fastest growth rate, the highest rate of tissue turnover,

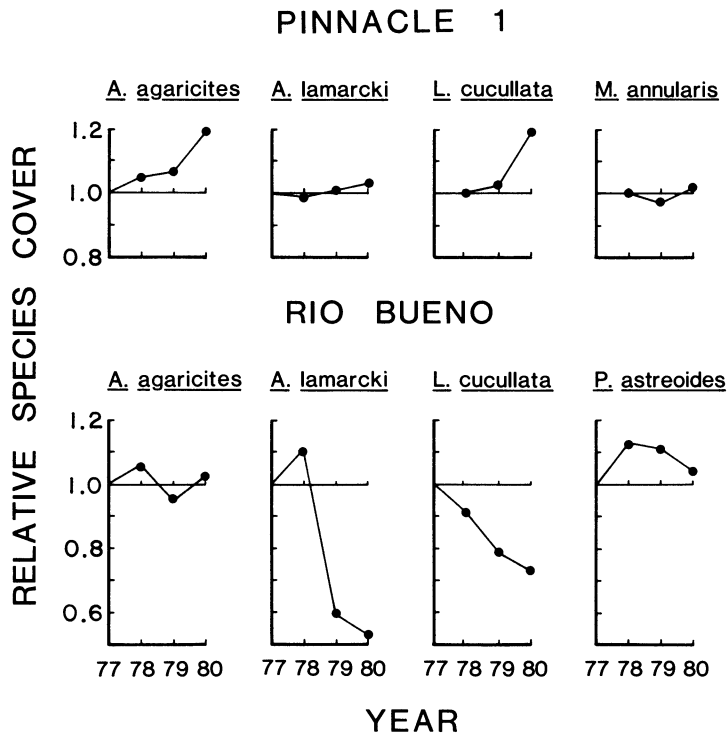


FIG. 11. Cover from 1977 through 1980 by each population (relative to area in 1977). Mean absolute percent cover is given in Table 2.

and the second highest rate of larval recruitment. Moreover, judging from the degree of early mortality (Table 6), the total larval influx, including settlement which went undetected between censuses, may well be the highest of the five species. In contrast, *Montastrea annularis* had the smallest proportion of small colonies, the slowest growth rate, the lowest mortality rates, the slowest rate of tissue turnover, and no larval recruits. Interestingly, these two species are also at op-

posite ends of a morphological sequence (T. P. Hughes, *personal observation*), in that *L. cucullata* is fragile and thin, while *M. annularis* plates are thick and robust. The other three species were intermediate between these two extremes, with *A. agaricites* and *P. astreoides* closer to *Leptoseris*, and *A. lamarcki* closer to *Montastrea*. Table 10 also suggests several correlations between the different life history parameters. Folioseous species with high rates of whole and partial mortality also have the heaviest larval settlement, the fastest individual growth rates, and the fastest rate of tissue turnover. From this, we predict that *L. cucullata* and *A. agaricites* are the best adapted for invasion of newly opened substrate, e.g., following a severe hurricane. Individual colonies of these two species settle in large numbers and grow quickly, and although they suffer heavy tissue losses, their populations also have extraordinary regenerative abilities (Table 9). In contrast, folioseous *M. annularis* and *A. lamarcki* should invade more slowly, but in areas of low disturbance may ultimately become abundant and long lived.

TABLE 9. Comparative regeneration capabilities of the different coral populations. The ability to recover lost tissue was measured crudely as change in coral area from 1977 to 1980 (column 1) minus the amount of live coral destroyed over that time (column 2) (see Table 3 and Fig. 5). The resulting area regenerated is given in column 3. All areas are percentages, relative to those in 1977.

Coral species	% change in coral area 1977-1980	% of coral lost from 1977-1980	% area regenerated
Pinnacle 1			
<i>A. agaricites</i>	+19	-33	52
<i>A. lamarcki</i>	+3	-9	12
<i>L. cucullata</i>	+19	-56	75
<i>M. annularis</i>	+2	-12	14
Rio Bueno			
<i>A. agaricites</i>	+3	-47	50
<i>A. lamarcki</i>	-47	-57	10
<i>L. cucullata</i>	-37	-103	66
<i>P. astreoides</i>	+4	-48	52

DISCUSSION

*Population dynamics of clonal organisms;
general considerations*

Many of the familiar demographic concepts developed in studies of aclonal organisms cannot be easily applied to clonal animals and plants. Conventional life history theory ignores the ecological and evolutionary

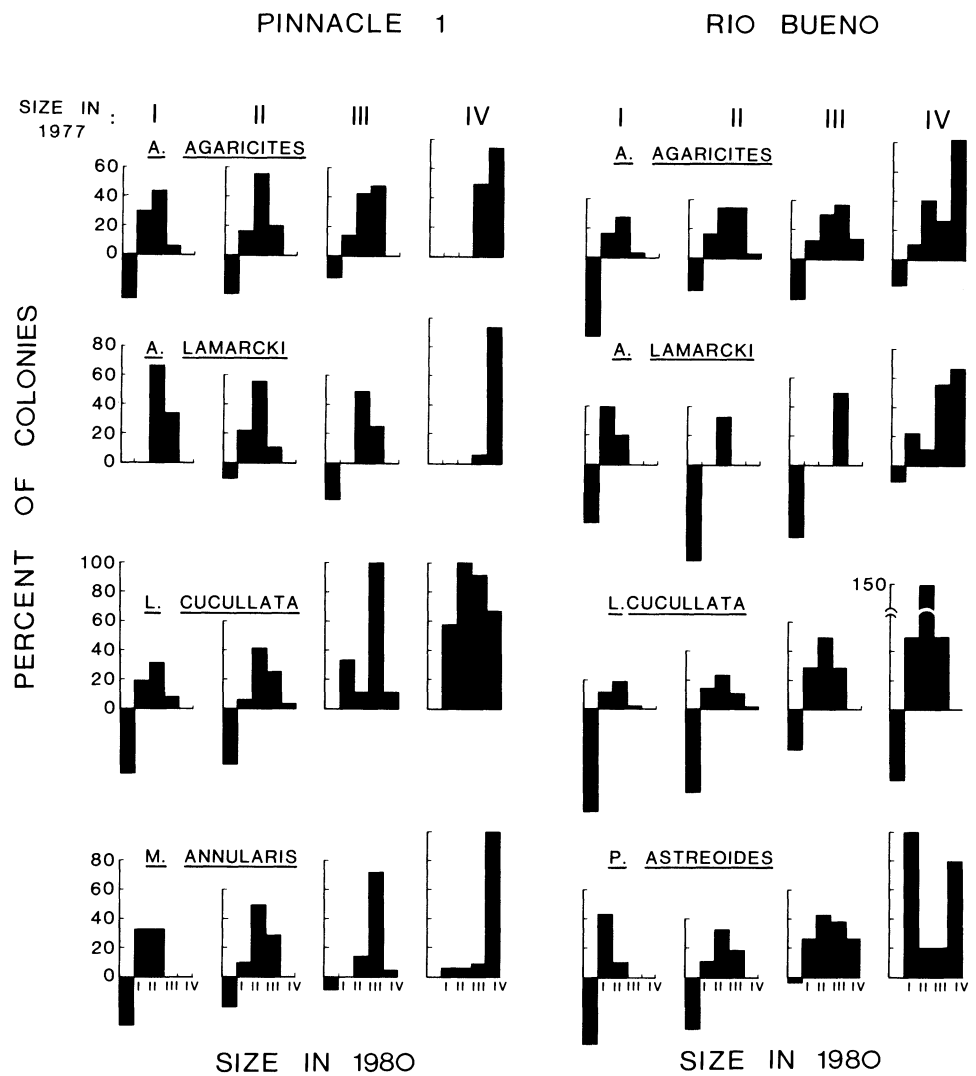


FIG. 12. Size-frequency distributions of colonies in 1980 as a function of their size 3 yr earlier. Percentages add up to >100% if the number of colonies generated by fission between 1977 and 1980 exceeds losses from mortality. Bars below the abscissa represent colonies alive in 1977 that died completely during the period (i.e., not including deaths of known daughter colonies). Bars above the abscissa represent size-classes of survivors.

consequences of modular processes such as partial mortality, fission, and fusion. Because of differential rates of growth and shrinkage, reproductive output and risk of mortality in clonal (and many aclonal) organisms are often dependent on size as well as age. We have shown that many of the largest (and therefore old) coral colonies in 1977, were reduced in area by >90% by 1980 (Fig. 12), while many juvenile corals (observed larval recruits) had increased in size, often far beyond these shrunken “adults.” Size and age interact in a complex fashion to determine the probability of a coral being injured or killed (Fig. 3), of undergoing fission (Table 4), or of growing or shrinking (Fig. 12). A large colony that shrinks will have an increased probability of mortality and decreased fecundity, so that it may resemble demographically other small colonies, what-

ever their age, more than its contemporaries that have remained large. Thus, many fundamental concepts such as age at first reproduction, age-specific fecundity and mortality, reproductive value, and senescence are difficult to relate to most clonal organisms.

From a practical point of view, to measure the age structure of a coral population would require following all individuals through a complete life cycle, or examining skeletal banding patterns in hundreds of colonies collected without bias. For foliaceous corals, the first method would take several human lifetimes, and the second would require considerable reef destruction. Not surprisingly, neither has been done for any coral species (but for excellent beginnings, see Connell 1973, and Vaisnys and Dodge 1980). Instead it is standard practice to estimate age by dividing colony diameter

TABLE 10. A summary of demographic and life history characteristics of the different coral species presented as rank values, 1 (highest) to 5 (lowest).

Characteristic	<i>A.</i> <i>agaricites</i>	<i>A.</i> <i>lamarcki</i>	<i>L.</i> <i>cucullata</i>	<i>M.</i> <i>annularis</i>	<i>P.</i> <i>astreoides</i>	Reference
	Rank of species					
Larval recruitment rate	1	4	2	5	3	Table 6
Growth rate	3	4	1	5	2	Figs. 9, 10
Proportion small colonies	2	4	1	5	3	Figs. 1, 2
Whole-colony mortality	2	3½	1	5	3½	Fig. 3
Partial mortality	3½	3½	1	5	2	Fig. 3
Tissue loss	2½	3	1	5	2½	Table 3
Population regeneration rate	2½	4½	1	4½	2½	Table 9
Mean rank	2.4	3.8	1.1	4.9	2.6	

by the average annual growth rate, assuming that variations in growth rate between colonies and between years are minimal (reviewed in Connell 1973). Yet our data indicate that partial colony mortality, fission, and fusion largely destroy the relationship between colony size and age in these foliaceous corals, and there is insufficient quantitative data available to test these assumptions for other growth forms (Connell 1973).

Of course, models of population dynamics can also be based on changes in individual (colony) size, rather than on age-dependent parameters. Transition probabilities from one size-class to another (e.g., Fig. 12) can readily be measured under natural conditions without the necessity of continual observations over entire generations (Lefkovitch 1965, Sarukhan and Gadgil 1974, Hartshorn 1975, Werner and Caswell 1977, Enright and Ogden 1979, Hughes 1984). The resulting matrix of probabilities is more than a tidy sum of interacting processes; it may also be used to predict population growth (Leslie 1945). Computer simulations of the dynamics of coral populations that behave by the rules of empirically determined (i.e., real-life) matrices can be used to examine the effects of modular and whole-colony processes on long-term individual, population, and community development (Hughes 1984).

Coral demography

Several of the life history patterns we have observed have been described from elsewhere in the Caribbean. In Curacao and Bonaire, abundance patterns of corals <4 cm in diameter are similar to those of the smallest size-class at our site (Bak and Engel 1979). As in Jamaica (Rylaarsdam 1983, and Fig. 1) and Florida (Dustan 1977), many of the most abundant coral species, in terms of their amount of cover, had very few small colonies. Also in Curacao (Bak and Luckhurst 1980), plates of *A. lamarcki* and *M. annularis* >30 cm in diameter had similar mortality patterns to each other, which were 4–6 times less than for *A. agaricites*. Since Curacao and Jamaica are more than 1200 km apart, it seems likely that the life history characteristics of the various foliaceous species remain approximately the same over a wide geographical area.

The most striking aspect of the demography of the

corals is its complexity. Otherwise, the range of life histories shown by the five coral species studied is considerable, but no greater than that observed within other clonal groups such as bryozoans (Winston and Jackson 1984) and sponges (Dayton 1979, Ayling 1980), or among aclonal animals. In comparison with *M. annularis*, *L. cucullata* is certainly a weedy or fugitive species, but many other epibenthic reef animals are much more ephemeral (Jackson 1977). In general, the shortest lived animals at our study sites are aclonal and/or smaller than foliaceous corals; they settle sooner and in much larger numbers, and they grow faster and die younger. There are, however, several implications of the complexity of clonal organism demography that deserve special mention, as discussed below.

Mortality patterns

Mortality was strongly dependent on colony size (Figs. 3 and 4). Connell (1973) suggested that the survivorship of small colonies is low because of their poor ability to recover from injuries. He hypothesized that smaller colonies have fewer reserves of energy or material to transfer from undamaged polyps for repairing wounds. Thus, regeneration should be slower and less successful, leading ultimately to complete mortality of the colony. In addition, small colonies probably are more susceptible to “instantaneous” whole-colony mortality from sources such as “bulldozing” by grazers or smothering by sedimentation (Bak and Engel 1979, Sammarco 1980, Rylaarsdam 1983). Jackson (1979) predicted that, on probabilistic grounds alone, the chance of a disturbance event affecting a colony increases with its surface area, but so too does the probability that some portion of the colony will not be affected. Therefore, small corals should escape or die completely, while larger colonies should seldom escape without suffering some amount of partial mortality (Hughes and Jackson 1980, and Fig. 3). Connell’s and Jackson’s hypotheses are not incompatible; colony size-dependent regeneration capability will enhance the differences predicted from probability considerations alone.

The major change in biomass in these populations caused by the storm of 1980 occurred independently

of any significant change in local population size (i.e., number of colonies), because of partial mortality. Although the total loss of coral tissue at Rio Bueno nearly doubled in the storm, to one-third of the combined populations' area, the annual loss of colonies was similar throughout the study period. Furthermore, while there was considerable mortality of entire large corals during the storm, the amount of tissue lost by partial colony mortality still remained greater for most species (Fig. 5). Clonality enabled individuals, or parts of them, to survive severe disturbances, thus effectively buffering fluctuations in gene frequency. This strongly contrasts with the effect of disasters on those aclonal organisms which cannot survive large amounts of injury (Coe 1957, Carson 1968, 1975). Note also that whether the effects of the storm are considered density dependent or independent is not determined by the number of individuals killed (the usual measure of population density), or by their combined area (e.g., Porter et al. 1981), but by the coral population structure. This is because storm mortality is size selective. A population dominated by large colonies will be severely damaged during a storm, while a population composed only of small individuals will show relatively little change in mortality patterns from calm to stormy years (e.g., compare *A. lamarcki* to *L. cucullata* at Rio Bueno in Fig. 5, see also Hughes 1984).

Even with much greater disturbances, such as those caused by hurricanes, large corals frequently survive as remnants or fragments (Glynn et al. 1964, Shinn 1972, Fishelson 1973, Stoddart 1974, Loya 1976c, Tunnicliffe 1980, Woodley et al. 1981, Highsmith 1982, and Fig. 5). Recovery of damaged reefs is much faster when some colonies survive and can regenerate, compared to the long-term effect of complete devastation, where reef recovery is dependent entirely on new sexual recruits (e.g., Shinn 1972, Endean and Stablum 1973, Stoddart 1974, Pearson 1981).

Fission and fusion

When a coral undergoes fission to produce a fragmented phenotype (sensu Harper 1977), its fitness presumably depends on the success of all of its component daughter colonies. Fission in these foliaceous corals is not initiated by the development of genetically determined (asexual) reproductive structures or tissues, as seen for example in budding of *Hydra*, or asexual reproduction of fungiid corals (Wells 1966). Instead, fission occurs as a consequence of partial mortality. Thus, the production of daughter colonies is associated with an immediate cost due to the loss of tissue killed during fission. Whether clones of daughter colonies ultimately gain from being separated depends on their combined survivorship (Cook 1979, 1983, Highsmith 1982), and on total sexual reproductive output of the genet (Hughes 1984). By producing daughter colonies, large foliaceous corals may lessen the danger of genet mortality from

rare disasters that are size selective. For example, several size-class IV plates were torn off the wall at Rio Bueno during the severe storm in March 1979, but almost half of these corals were survived by small remnants formed by colony fission a year earlier. Fission could also decrease genet mortality due to infectious disease or parasites. Although fission may be a disadvantage during routine years, over longer time scales it may actually increase genet fitness.

Apart from this study (Tables 4 and 6), the comparative survivorship of sexually and asexually derived corals is unknown. There are also no other quantitative measurements of frequencies of coral fission (Fig. 7, Table 4). Nevertheless, it is apparent that fission in foliaceous corals is relatively unimportant compared to fission and fragmentation in some ramose species, especially many loosely branching acroporids. Larval recruitment is very rare in many (but by no means all) ramose species, which apparently rely heavily for propagation on asexual fragmentation and dispersal by wave action (e.g., Gilmore and Hall 1976, Tunnicliffe 1980, Bothwell 1981, Highsmith 1982, Rylaarsdam 1983).

The repair of damaged tissue immediately after fission seems to have resulted in much of the colony growth taking place preferentially at the site of injury (see also Stephenson and Stephenson 1933, Kawaguti 1937, Stephenson et al. 1958, Loya 1976a). For plating corals, this directed growth tends to bring fragments back together, although it may take many years before they touch. In all such cases of renewed contact we observed, known daughter colonies fused to form a single colony again, although occasionally, small daughter colonies were overtopped and killed by larger fragments of themselves (i.e., intragenet competition).

The genetic diversity of coral populations

Because of fissions which undoubtedly occurred before our study began we do not know how many of the original 883 colonies were genetically distinct (i.e., were genets, sensu Harper 1977). The fusion data suggest that the number of genets was at least 40 less, or 843. If we set up boundary conditions by making the unlikely assumption that all these 843 colonies were of different genotype, we can calculate the absolute minimum by which counts of colonies in 1980 exceeded the number of genets. Because of the average net annual input of 43 asexually formed colonies, the number of colonies in 1980 was at the very least 19% greater. Clearly, genetic mapping by fusion experiments (e.g., Potts 1976, Bothwell 1981, Neigel and Avise 1983) or electrophoresis would show a much greater discrepancy.

Harper (1977) argued that only modules and genets need to be considered in population studies of clonal organisms, since neither fission, fusion, nor the death of daughter colonies cause any change in genetic di-

versity. While that is obviously true, we have shown that the manner in which a genet's modules are distributed in space, either as single or multiple colonies, will affect its overall survival (and probably also fecundity), as well as obviously influencing population size structure. The production and loss of daughter colonies means that there is a flux in the number of physiologically separate, countable individuals, and they must be taken into account.

Coral growth

We have shown that over periods of a few months to years, increases in colony area, or any other measures of growth, are likely to be erratic, because sooner or later the growth of a coral will be interrupted or reversed by external factors such as sedimentation, competition, or predation (Fig. 3). Indeed, in the 3 yr of our study, only 16.2% of the monitored colonies escaped without being injured at least once. It is therefore essential to distinguish between the inherent potential for individual colony growth and realized net growth (Bak 1976). Predictions of colony growth rates extrapolated from short-term calcification measurements (often taken over only a few days or weeks) are often invalidated by growth reversals.

The present study confirms earlier indications that species growth rates do sometimes, on average, slow down with increasing depth (e.g., Buddemeier et al. 1974, Baker and Weber 1975, Dustan 1975, Bak 1976, Gladfelter et al. 1978, Highsmith 1979, Tunnicliffe 1980), but the range in values among colonies at any depth is often considerable. Obviously, statements comparing growth rates at different depths or sites are or ought to be statistical rather than absolute. Neither depth nor environment has been shown to be a particularly reliable predictor of coral growth in this or any other study.

The decline in relative growth rate as a function of coral size (Fig. 10) suggests that there are strong selective pressures acting to maintain the growth rate of small colonies at high levels, so that they can reach a size refuge from mortality (Fig. 4). This is indicated further by the continued rapid growth rates in cohort II of *Leptoseris cucullata* and *Agaricia agaricites* (Fig. 10), the two species with the highest rates of small colony mortality.

Whether or not corals of identical size but with different histories of growth and shrinkage have similar reproductive outputs is unknown. Clearly, their past (and probably future) contributions to the gene pool will be very different (Hughes 1984). Both Stimson (1978) and Rinkevich and Loya (1979a) have reported that reproductive tissues are located towards the center of coral colonies. Perhaps this is because many agents of partial colony mortality often work from the edge of a colony inwards. By locating gonads centrally, corals could often avoid, or at least postpone, loss of reproductively valuable tissues.

Longevity and senescence

The lack of correlation between colony size and linear extension rate indicates that growth is genetically indeterminate in the five species examined here, and there is no evidence for senescence in foliaceous or any other corals (Connell 1973, Highsmith 1979). Many of the largest colonies measured (Table 10) were daughter colonies surrounded by bits and pieces of themselves, i.e., they formed part of a clone of colonies. In the case of deepwater *A. lamarcki* and *M. annularis*, the genet as a whole can be up to 5 m across, so that even if they grew at a sustained rate of twice the average (Fig. 9), these genetic individuals are many centuries old. Once colonies reach a moderately large size, the probability of their being killed falls almost to zero. Indeed, not a single colony of *M. annularis* and *A. lamarcki* in size-class III and IV at Pinnacle 1 has been killed from 1977 to the present (1983), a period of time which included a major storm and a hurricane. For *A. agaricites*, *L. cucullata*, and *P. astreoides*, the period between hurricanes is almost certainly longer than the life-span of a genet, but for deepwater *M. annularis* and *A. lamarcki*, rare disasters such as major hurricanes once every 100 yr are predictable events, likely to be experienced several times and therefore significant selective factors in these corals' evolution (cf. Slobodkin 1968). In this case, the most important unpredictable event in these corals' lives may be major changes in sea level associated with the Pleistocene glaciations (Potts 1983, 1984).

Coral life histories and community organization

One important distinction among clonal species, arising from differences in their life histories and morphology, is their ability to extend across the substratum (Buss 1979, Jackson 1979). Although all sessile organisms are, by definition, incapable of active movement, a clonal animal or plant may nevertheless sample new microhabitats by growth and by passive dispersal of fragments. Different species vary enormously in the extent of this "mobility" (Jackson and Hughes 1985). For example, *L. cucullata* rapidly produces a thin, fragile skeleton, and during its lifetime a colony may meander over the reef surface, often several metres away from where it originally settled. Nevertheless, *Leptoseris* loses in most of its direct competitive encounters with other clonal species (Lang 1973), and does not persist in any one place for long (cf. Table 9). In contrast, foliaceous *M. annularis* is much more stationary. It grows laterally 5–10 times slower than *L. cucullata*, is very sturdily constructed, and may occupy the same space for many years. *Montastrea* also repairs wounds in its tissue and skeleton faster than any other species of plating coral (Bak et al. 1977, Bak and Van Es 1980), and is a good competitor, capable of digesting all other foliaceous corals with which it comes into contact (Lang

1973). However, as noted earlier, larval recruitment by *Montastrea* is far less than for *Leptoseris* (Bak and Engel 1979, Rylaarsdam 1983; Table 6).

Differences in life history and mobility between species are reflected in the taxonomic and morphological composition of coral communities over the reef. Among foliaceous corals, only more mobile species like *L. cucullata*, *A. agaricites*, and *P. astreoides* are common on shallow reefs (Goreau and Wells 1967) where disturbance from storms and grazing predators is greatest (e.g., Adey 1978, Bak and Luckhurst 1980, Woodley et al. 1981, Jackson and Winston 1982). In contrast, the more stationary foliaceous colonies of *M. annularis*, *A. lamarcki*, and *A. grahamae* become dominant only below ≈ -20 m, where sponges, which are fierce competitors with corals, are most abundant (Reiswig 1973, Suchanek et al. 1983). In spite of their more fragile construction, highly mobile foliaceous species are often favored over more stationary ones where the chances (or duration) of persistence are lessened by disturbance.

Mobile and stationary life histories are also found among other morphological groups, such as branching and massive corals. Loosely branching species are often extremely mobile, and a single clone can extend up to 10 m in diameter through a repeated sequence of rapid growth, breakage, and passive tumbling (Gilmore and Hall 1976, Tunnicliffe 1981, Bak and Crieis 1982, Highsmith 1982, Neigel and Avice 1983). As described earlier, several fragments of ramose *Acropora cervicornis* and *Porites furcata* were established in this way within some of the shallower (-10 -m) quadrats at Rio Bueno. In comparison, massive colonies, such as those of brain corals, are generally more stationary than branching or foliaceous forms. They grow slowly (Connell 1973), are much superior competitors in direct encounters with other corals (Lang 1973), and are less susceptible to storm damage (Woodley et al. 1981). Of course, there is considerable variation within each of these morphological groups, as we have shown here for foliaceous corals.

In the Caribbean, branching acroporids are typically the most abundant corals at or below the reef crest (Goreau and Wells 1967, Adey 1978), where they must withstand considerable amounts of wave action. Although massive corals are inherently much more resistant to damage from waves, they are nevertheless often found slightly deeper (3–12 m), below the most common branching corals, particularly the elkhorn coral, *Acropora palmata*. Many branching species such as *A. palmata* and the staghorn coral, *A. cervicornis*, appear to rely heavily on fragmentation and passive dispersal during storms for propagation, rather than on sexual recruitment (e.g., Gilmore and Hall 1976, Bothwell 1981, Tunnicliffe 1981, Highsmith 1982). Thus, in addition to strongly influencing the community structure, zonation, and diversity of coral reefs (Connell and Slatyer 1977, Connell 1979, Woodley et

al. 1981), disturbance has strongly influenced the evolution of different life histories among corals and other reef organisms (Karlson 1980, Jackson and Hughes 1985).

Although many mobile species of corals are well adapted to moderate levels of damage, there is an upper limit to their tolerance of disturbance. For example, *A. palmata* and *A. cervicornis* are badly damaged by severe hurricanes (Woodley et al. 1981, Rogers et al. 1982), from which they recover only slowly (Knowlton et al. 1981). Few corals can become established in shallow water on the highly exposed reefs close to storm tracks, where the reef crest is constructed instead of crustose algae (Adey 1978). Foliaceous corals are also relatively rare on shallow or exposed reefs. Interestingly, several species of corals, notably *M. annularis*, *A. agaricites*, and *P. astreoides* change from a foliaceous to a mound-like morphology in shallow water (Dustan 1975). This morphological transition is gradual, but the changeover occurs at ≈ -25 to -30 m in stationary *M. annularis*, and at -8 to -15 m in the two more mobile species. The most mobile foliaceous species, *L. cucullata*, can persist with a plate-like morphology even as shallow as -3 m (Goreau and Wells 1967).

Along with disturbance from grazers and storms, competition for space is also important in structuring coral reef communities, and has been described from all reef zones (e.g., Lang 1973, Jackson 1977, Suchanek et al. 1983, and many more). In particular, descriptions of coral-coral competition through overtopping (Porter 1974, Connell 1976) and by aggressive extra-coelenteric feeding (Lang 1973) are common in the literature, although their importance will remain unknown until the relative frequencies of mortality from all sources are documented. Physiological tolerance, especially in relation to light, may also have an important role in coral zonation, e.g., the deepest reef-building corals, below ≈ -50 m, are usually very fragile foliaceous species, such as *Agaricia fragilis* and *A. undata* (Goreau and Wells 1967). The replacement of branching and massive corals by foliaceous forms in deeper water has traditionally been interpreted as a response to depth-related gradients in light, although, as we have argued above, many other factors are involved.

Clonal plants also show striking variation in their mobility and life histories, which have been extensively studied in relation to terrestrial succession and community organization (e.g., Watt 1947, Sarukhan and Gadgil 1974, Grubb 1977, Harper 1977, Cook 1983). Lovett Doust (1981) proposed the term guerilla to characterize loosely connected, mobile species, and phalanx to describe stationary species with tightly clumped modules. As the names imply, plants with a guerilla strategy can grow rapidly, but are poor competitors, while the opposite is true for phalanx species. Just as with corals on reefs, mobile plant clones are generally

associated with highly disturbed habitats, such as fire-prone grasslands, and are among the first to invade space newly opened by disturbance (Kerster 1968, Platt 1975, Turkington and Harper 1979).

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